

DI FA SEA TL

Feature Specification

Project / Product	
Project name / version	Technological functions
Product name / version	Feature Specification for TM FAST V2.0
Project ID	brh-1430 / brh-1550
PA number	2022
Product family	ET 200MP
Path to PNB	\\pnb04.siemens.net\PNB_TM_CM\S7-1500\smartTM_554-1AA00-0AB0\000_Update\FW_V2.0

Document	
Document version	2.1
Document status	Released
Document date	21.02.2025
Document number TC	
Protection class	Confidential
Author	Wild, Guillaume (DI FA SEA TL PM, guillaume.wild@siemens.com)
File name	Feature_Specification_TM_FAST_V2.1.docm

Review:

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Change Log:

Version	Date	Author	Status	Change Comment
1.99	05.02.2025	Wild		First review comments integrated
2.0	18.02.2025	Wild	Released	All inputs from both reviews integrated
2.1	21.02.2025	Wild	Released	Only change in parametrization record (offset of the 2 parameters UserProcessingTimeInput and UserProcessingTimeOutput changed)

General hint:

Texts in blue are usually instructions for completion or suggestions. Please adapt your entries according to your requirements and delete blue texts.

To release the document, no more blue text should be contained.

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1 Scope

The scope of this document is to gather internal knowledge about the design and usage of the TM FAST module. Even if the main purpose of this document is the firmware design, an overview of customer target aspects, hardware design, engineering and software is provided as well. Not contained in this document are the ET 200MP common system features.

2 General

Machines and processes in the automated world are getting faster and faster. At the same time the required accuracy rises. In the S7-300 system we offered the High-speed Boolean processor FM 352-5 which allows the user to program the HW functionality inside a function module by loading his own FPGA program to the module. FPGA stands for "Field Programmable Gate Array" and is a kind of ASIC where the physical and logical interconnections can be loaded. This brought SIMATIC in a unique position of fast reaction applications. In S7-1500 or ET 200SP portfolio a comparable module is not available yet and therefore still a migration impediment.

Typical requirements that are common for many users are solved with our standard technology modules like the TM Count for a wide range of counting applications. As soon as more special requirements appear it's not always sensible to solve it with a standard module but give the customer the chance to solve it by himself.

In the market there is nearly no comparable offering for the applications solved with our High-speed Boolean processor FM 352-5 and customers are asking for a future proof solution in the SIMATIC S7-1500 automation system.

Our technology modules in SIMATIC S7-1500 automation system consist of a backplane with a programmable FPGA combined with technology specific connection PCBs (function boards) for IOs. This gives us a perfect HW platform for opening the FPGA to user access and so open the opportunity for application specific, user defined technology modules.

Therefore, we want to offer a module where our customers can design their own functionality in our HW base.

Thinkable tasks may be:

- counting pulses, acquisition logics for incremental and absolute encoders and deriving any immediate reaction based on other circumstances
- generating pulses and patterns at digital outputs with high precision and resolution far below 1 μ s
- connecting digital signals in any user defined combination and deriving reactions to digital outputs with defined timing conditions
- reading in any digital signal and analyzing it in his timing conditions
- performing calculations and preprocessing within the module for example for condition monitoring and early defect detection

For more details about the requirements & motivation, see [/Requirements/](#)

2.1 Brief Description

The TM FAST module is giving the customer a frame to embed its very own application into the SIMATIC S7-1500 automation system.

From the hardware point-of-view the module is made of a backend device (Siemens PER4-1 ASIC) hosting the Siemens TM FAST firmware and a frontend device (Intel Cyclone10 LP FPGA) that will host a so-called programmable "TMFASTApplication". This "TMFASTApplication" is generated by a Quartus project which is linking a Siemens FAST-system logic and a customer specific user logic.

To design its own application-specific TM FAST-module, the customer will first design its own FPGA "user logic" using provided examples of user logics including ModelSim simulation projects, a prepared Quartus project and the migrated logic library of the Boolean Processor FM352-5. To support the development of customer specific applications, a TM FAST debug connector will be available to directly program and debug the frontend device (FPGA) over the Intel FPGA Download Cable (USB-Blaster). The debugger interface is by default blocked (not for productive use) and can be activated using the TMFASTControlREC.

The resulting "TMFASTApplication" will be loaded onto the module and finally the module will run the application and make it accessible to the CPU and process side.

For using all debugging features of the TM FAST, USB-Blaster (Intel FPGA Download Cable) is required and will have to be purchased directly by the customer. Note that the debug connector as shown below is reducing the ventilation capability of the ET200MP TM FAST and therefore should not be kept permanently plugged on the module in industrial operation.

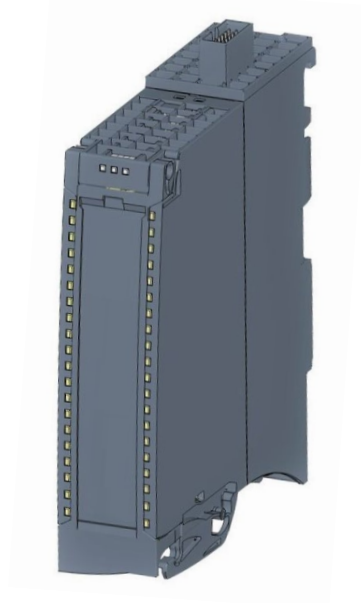


Figure 1: Picture TM FAST with Debug Connector plugged

2.2 Product Structure

2.2.1 TM FAST block diagram

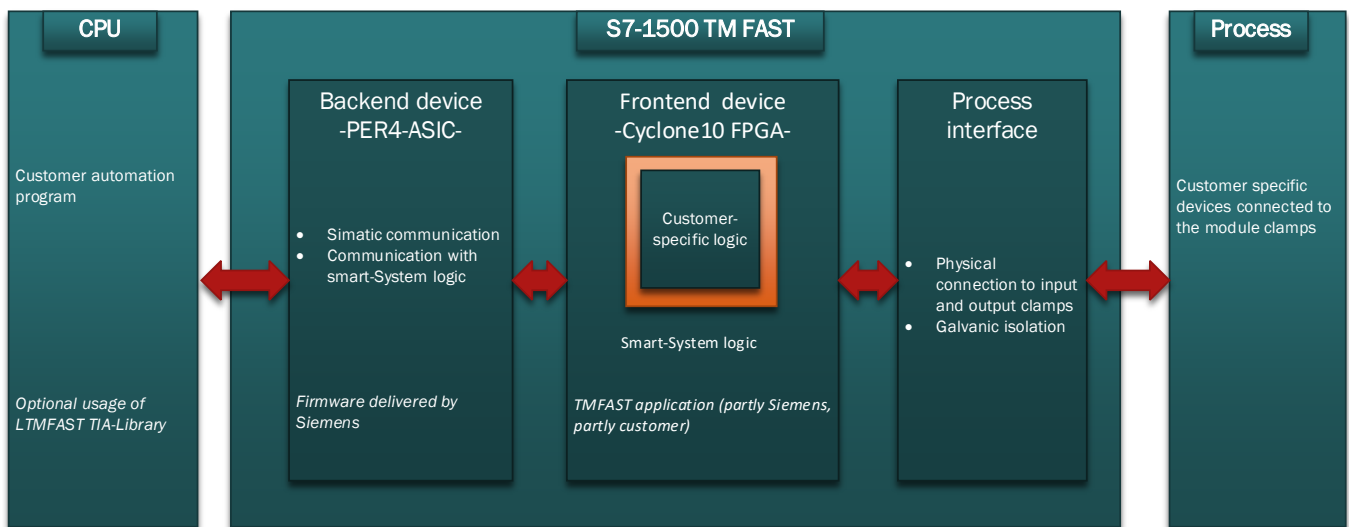


Figure 1: TM FAST Principle

2.2.2 TM FAST backend device (PER4-1 ASIC)

The backend device is realizing the connection of the TM FAST to the SIMATIC-world over the backplane bus of the SIMATIC S7-1500 automation system (PBus+). The firmware running on the backend device is delivered by Siemens - signed update file, like standard SIMATIC S7-1500 modules.

The following main tasks are running within the backend firmware:

- PBus+ handling: communication with the S7-1500 head station, cyclic (IO-data exchange) and acyclic communication (records and diagnostics exchange)
- loading the TMFASTApplication from the flash device into the FPGA over SPI-Interface
- validation of the loaded TMFASTApplication over CtC (chip-to-chip) interface
- activation of the TMFASTApplication upon request
- detection of any failure of the frontend device
- setup the FAST system logic according to the module parametrization
- accesses as a master from the frontend device must be blocked: CtC page settings must ensure that the TMFASTApplication cannot get direct access to the SIMATIC components
- update flash content (firmware-update or TMFASTApplication download)
- retrieve temperature measurement from temperature sensor over I2C
- control all LEDs of the TM FAST module over SPI (LED-drivers are located on the function boards)
- support for module manufacturing (code is integrated in the PER4-1 ASIC integrated secondary bootloader)
- checking the soft error detection state of the FPGA

2.2.3 TM FAST frontend device (Cyclone10 LP FPGA)

The frontend device is the core of the TM FAST where the customer-specific application is located. The FPGA program (TMFASTApplication) is containing two parts: the FAST system logic (TFL_FAST_SYS) and the customer-specific user logic (TFL_FAST_USER). The customer's design is embedded in a shell which interface is described in a so-called VHDL entity file TFL_FAST_USER_e.vhd and visible as well in the file TFL_FAST_USER_e.bsf, the Quartus block symbol file for the top entity.

2.2.3.1 Description TM FAST system logic (TFL_FAST_SYS)

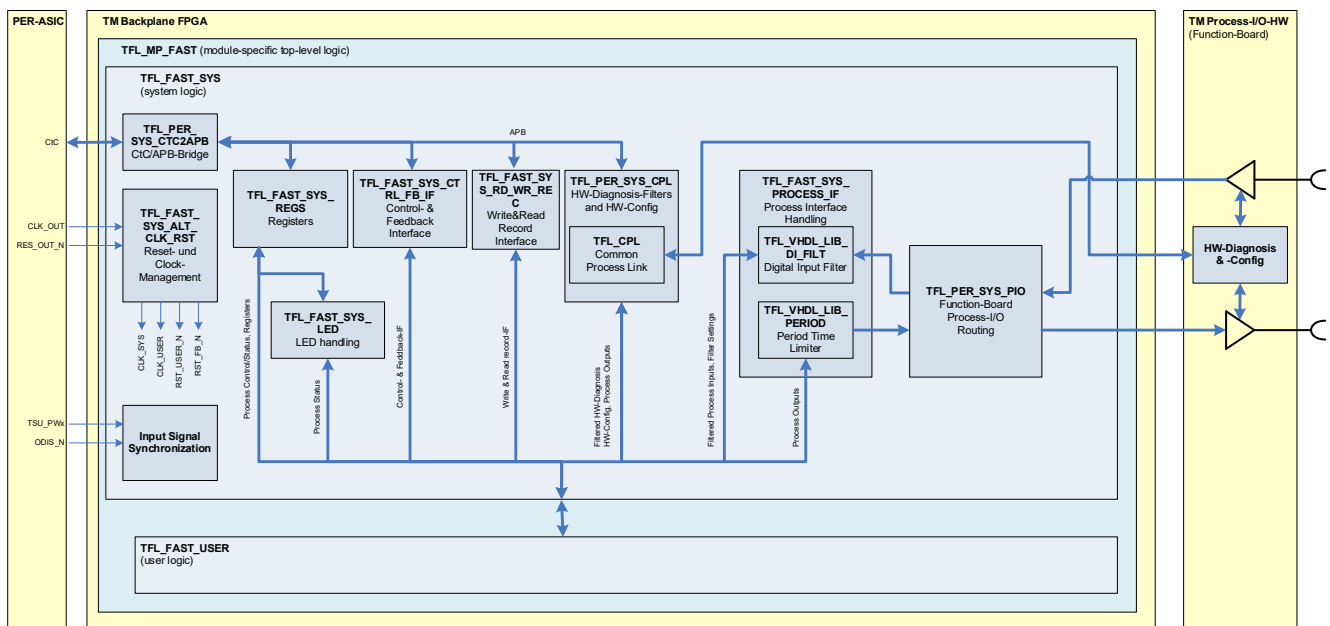


Figure 2: FPGA design overview

2.2.3.2 Description of the user configuration

The customer shall configure the Siemens system logic by setting some constants in the file TFL_FAST_USER_IP_CONF_PUBLIC_p.vhd.

The following parameters of the system logic can be setup:

- ➔ **USER_LOGIC_VERSION**: user logic version number, used format <letter><xx>.<yy>.<zz> with letter R,B,V,P,T and digits xx,yy,zz are decimal values between "00" and "99". **Note that the firmware V1.0.0 is not accepting a V-Version with xx equal to 0.**

- ➔ **APPLICATION_ID**: 8 characters describing the TMFASTApplication, letters and digits shall be used, starting characters must be a letter.
- ➔ **FUNCTION_ID**: license ID provided by Siemens / Wibu-Systems, 0 if no licensing is required.
- ➔ **WR_REC_DWORD_SIZE**: number of double-words within the user write record, allowed are the following values only: 1,8,16,32 – one double-word is made of 4 Bytes.
- ➔ **RD_REC_DWORD_SIZE**: number of double-words within the user read record, allowed are the following values only: 1,8,16,32 – one double-word is made of 4 Bytes.
- ➔ **F_CLK_USER**: user clock frequency – allowed values are 5_000_000 (5MHz), 15_000_000 (15MHz), 25_000_000 (25MHz), 50_000_000 (50MHz), 75_000_000 (75MHz).
- ➔ **PHASE_QUANTITY**: quantity of user clock phases, used by Boolean library components, allowed are values between 0 (no phases) and 14 – only for customer migrating an application from Boolean processor FM352-5, for new designs recommended to keep the value 0 and not use any phases.
- ➔ **RS485_OR_TTL_DIRECTION**: mode of each RS485 channel, possible values are: unsued, RS422_input, RS422_output, TTL_input, TTL_output, RS485_bidir
- ➔ **RS485_TERMINATION**: termination state of one RS485 channel, possible values are: unavailable, on, off or parametrizable.
- ➔ **DI_FILTER_USER_VAL_MS**: user defined digital input filter time in millisecond, can be set separately for each digital input (range 0ms to 27.96ms, resolution of 13.33ns)
- ➔ **RS485_FILTER_USER_VAL_MS**: user defined RS485 input filter time in millisecond, can be set separately for each digital input (range 0ms to 27.96ms, resolution of 13.33ns)

2.2.3.3 Interface between TFL_FAST_SYS and TFL_FAST_USER

The entity is describing the interface that must be used by the user logic. The customer will implement its application as a behaviour (so-called VDHL architecture) within this fix entity.
Here is the description of the entity "TFL_FAST_USER_e".

2.2.3.3.1 Entity system logic V1.0

```

-----
-- top heading: Entity of User Logic of TM FAST
-- filename: TFL_FAST_USER_e.vhd
-- copyright: (c) Siemens AG, Automation Systems
-----

library ieee;
use ieee.std_logic_1164.all;
use work.TFL_FAST_USER_IP_CONF_p.all;
use work.TFL_FAST_USER_PUBLIC1_p.all;
use work.TFL_FAST_USER_PUBLIC2_p.all;
use work.TFL_FAST_USER_IP_CONF_PUBLIC_p.all;

entity TFL_FAST_USER_e is
generic (
    PARAM_INT_1: integer := 0;
    PARAM_REAL_1: real := 0.0;
    PARAM_BOOL_1: boolean := false;
    PARAM_STRING_1: string := ""
);
port (
    -- Logic-Clock and -Reset
    CLK: in std_logic; -- clock of user logic (low-to-high-edge!)
    PHASE: in std_logic_vector ( PHASE_MAX_QUANTITY downto 0 ); -- clock phases for boolean library components
    RST: in std_logic; -- reset of user logic (high-active!)

    -- Control and Feedback Interface
    CTRL_IF: in DWORD_VECTOR ( ( CTRL_IF_DWORD_SIZE - 1 ) downto 0 ); -- control interface
    FB_IF: buffer DWORD_VECTOR ( ( FB_IF_DWORD_SIZE - 1 ) downto 0 ); := -- feedback interface
    -- DWORD_VECTOR is defined in TFL_FAST_USER_PUBLIC2_p

    -- Read and Write Record
    WR_REC: in DWORD_VECTOR ( ( WR_REC_DWORD_MAX_SIZE - 1 ) downto 0 ); -- write record
    RD_REC: buffer DWORD_VECTOR ( ( RD_REC_DWORD_MAX_SIZE - 1 ) downto 0 ); := -- read record
    -- DWORD_VECTOR is defined in TFL_FAST_USER_PUBLIC2_p
    -- tic signal for new write record to user logic
    WR_REC_NEW: in std_logic;
    RD_REC_BUSY: out std_logic; -- tic signal for new read record from user logic

    -- Process I/Os
    DI: in std_logic_vector ( ( DI_QUANTITY - 1 ) downto 0 ); -- digital process inputs (Dis) (inputs are already filtered)
    DQ: buffer std_logic_vector ( ( DQ_QUANTITY - 1 ) downto 0 ); := -- digital process outputs (DQs)
    -- (others => '0');

    -- RS485 I/Os
    RS485_RX: in std_logic_vector ( ( RS485_QUANTITY - 1 ) downto 0 ); -- receive data
    RS485_TX: buffer std_logic_vector ( ( RS485_QUANTITY - 1 ) downto 0 ); := -- transmit data
    -- (others => '1');
    RS485_OE: buffer std_logic_vector ( ( RS485_QUANTITY - 1 ) downto 0 ); := -- transmit output enable
    -- (others => '0');

    -- quality informations (QI) 1=BAD 0=GOOD
    LP_QI_BAD: in std_logic; -- quality information of L+ supply
    DI_QI_BAD: in std_logic_vector ( ( DI_QUANTITY - 1 ) downto 0 ); -- quality information of digital process inputs (Dis)
    DQ_QI_BAD: in std_logic_vector ( ( DQ_QUANTITY - 1 ) downto 0 ); -- quality information of digital process outputs (DQs)
    RS485_QI_BAD: in std_logic_vector ( ( RS485_QUANTITY - 1 ) downto 0 ); -- quality information of RS485 channels

    -- SIMATIC CPU STOP
    CPU_STOP: in std_logic; -- SIMATIC CPU status 1=STOP 0=RUN

    -- Isochronous Signals
    T_DC: in std_logic; -- TDC time mark (high for only one CLK!)
    T_I: in std_logic; -- Ti time mark (high for only one CLK!)
    T_O: in std_logic; -- To time mark (high for only one CLK!)
);
end entity TFL_FAST_USER_e;

```

The following picture shows the schematic connections between the System-Logic and the User-Logic.

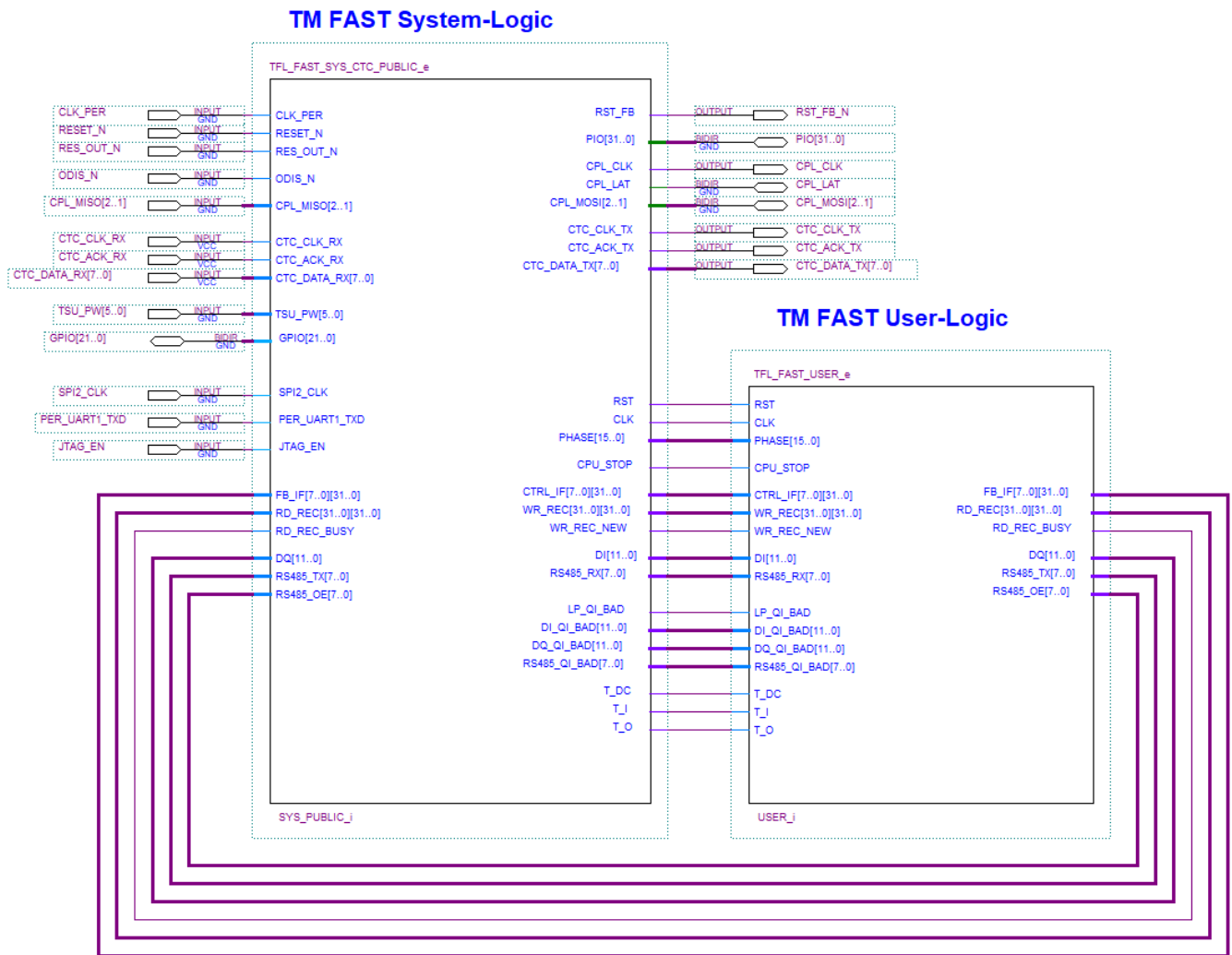


Figure 3: Logic design schematic top view (V1.0)

2.2.3.3.2 Entity system logic V2.0

The entity has been reworked for the V2.0 version of the system logic to support the Quartus schematic feature. From a functional point of view the only change is the addition of the isochronous signals T_I_LATCH and T_O_LATCH.

entity TFL_FAST_USER_e is

```
generic (
    PARAM_INT_1: integer := 0;
    PARAM_REAL_1: real := 0.0;
    PARAM_BOOL_1: boolean := false;
    PARAM_STRING_1: string := ""
);
port (
    -- Logic-Clock and -Reset
    CLK: in std_logic;
    PHASE: in std_logic_vector ( PHASE_MAX_QUANTITY downto 0 );
    RST: in std_logic;
    -- Control and Feedback Interface
    CTRL_IF0: in DWORD;
    CTRL_IF1: in DWORD;
    CTRL_IF2: in DWORD;
    CTRL_IF3: in DWORD;
    CTRL_IF4: in DWORD;
    CTRL_IF5: in DWORD;
    CTRL_IF6: in DWORD;
    CTRL_IF7: in DWORD;
    FB_IF0: buffer DWORD := ( others => '0' );
    FB_IF1: buffer DWORD := ( others => '0' );
    -- for internal use only, don't use the generics
    -- clock of user logic (low-to-high-edge!)
    -- clock phases for boolean library components
    -- reset of user logic (high-active!)
    -- control interface
    -- DWORD_VECTOR is defined in TFL_FAST_USER_PUBLIC2_p
    -- feedback interface
    -- DWORD_VECTOR is defined in TFL_FAST_USER_PUBLIC2_p

```

```

FB_IF2:    buffer DWORD := ( others => '0' );
FB_IF3:    buffer DWORD := ( others => '0' );
FB_IF4:    buffer DWORD := ( others => '0' );
FB_IF5:    buffer DWORD := ( others => '0' );
FB_IF6:    buffer DWORD := ( others => '0' );
FB_IF7:    buffer DWORD := ( others => '0' );
-- Read and Write Record
WR_REC00:  in    DWORD;
WR_REC01:  in    DWORD;
WR_REC02:  in    DWORD;
WR_REC03:  in    DWORD;
WR_REC04:  in    DWORD;
WR_REC05:  in    DWORD;
WR_REC06:  in    DWORD;
WR_REC07:  in    DWORD;
WR_REC08:  in    DWORD;
WR_REC09:  in    DWORD;
WR_REC10:  in    DWORD;
WR_REC11:  in    DWORD;
WR_REC12:  in    DWORD;
WR_REC13:  in    DWORD;
WR_REC14:  in    DWORD;
WR_REC15:  in    DWORD;
WR_REC16:  in    DWORD;
WR_REC17:  in    DWORD;
WR_REC18:  in    DWORD;
WR_REC19:  in    DWORD;
WR_REC20:  in    DWORD;
WR_REC21:  in    DWORD;
WR_REC22:  in    DWORD;
WR_REC23:  in    DWORD;
WR_REC24:  in    DWORD;
WR_REC25:  in    DWORD;
WR_REC26:  in    DWORD;
WR_REC27:  in    DWORD;
WR_REC28:  in    DWORD;
WR_REC29:  in    DWORD;
WR_REC30:  in    DWORD;
WR_REC31:  in    DWORD;
RD_REC00:  buffer DWORD := ( others => '0' );
RD_REC01:  buffer DWORD := ( others => '0' );
RD_REC02:  buffer DWORD := ( others => '0' );
RD_REC03:  buffer DWORD := ( others => '0' );
RD_REC04:  buffer DWORD := ( others => '0' );
RD_REC05:  buffer DWORD := ( others => '0' );
RD_REC06:  buffer DWORD := ( others => '0' );
RD_REC07:  buffer DWORD := ( others => '0' );
RD_REC08:  buffer DWORD := ( others => '0' );
RD_REC09:  buffer DWORD := ( others => '0' );
RD_REC10:  buffer DWORD := ( others => '0' );
RD_REC11:  buffer DWORD := ( others => '0' );
RD_REC12:  buffer DWORD := ( others => '0' );
RD_REC13:  buffer DWORD := ( others => '0' );
RD_REC14:  buffer DWORD := ( others => '0' );
RD_REC15:  buffer DWORD := ( others => '0' );
RD_REC16:  buffer DWORD := ( others => '0' );
RD_REC17:  buffer DWORD := ( others => '0' );
RD_REC18:  buffer DWORD := ( others => '0' );
RD_REC19:  buffer DWORD := ( others => '0' );
RD_REC20:  buffer DWORD := ( others => '0' );
RD_REC21:  buffer DWORD := ( others => '0' );
RD_REC22:  buffer DWORD := ( others => '0' );
RD_REC23:  buffer DWORD := ( others => '0' );
RD_REC24:  buffer DWORD := ( others => '0' );
RD_REC25:  buffer DWORD := ( others => '0' );
RD_REC26:  buffer DWORD := ( others => '0' );
RD_REC27:  buffer DWORD := ( others => '0' );
RD_REC28:  buffer DWORD := ( others => '0' );
RD_REC29:  buffer DWORD := ( others => '0' );
RD_REC30:  buffer DWORD := ( others => '0' );
RD_REC31:  buffer DWORD := ( others => '0' );
WR_REC_NEW: in    std_logic;
RD_REC_BUSY: out   std_logic;

-- write record
-- DWORD_VECTOR is defined in TFL_FAST_USER_PUBLIC2_p

-- read record
-- DWORD_VECTOR is defined in TFL_FAST_USER_PUBLIC2_p

-- tic signal for new write record to user logic
-- tic signal for new read record from user logic

-- Process I/Os
DI:      in    std_logic_vector ( ( DI_QUANTITY - 1 ) downto 0 );
DQ:      buffer std_logic_vector ( ( DQ_QUANTITY - 1 ) downto 0 ) :=
          ( others => '0' );
-- digital process inputs (DIs) (inputs are already filtered)
-- digital process outputs (DQs)

```

```
-- RS485 I/Os
RS485_RX: in std_logic_vector ( ( RS485_QUANTITY - 1 ) downto 0 ); -- receive data
RS485_TX: buffer std_logic_vector ( ( RS485_QUANTITY - 1 ) downto 0 ) := -- transmit data
           ( others => '1' );
RS485_OE: buffer std_logic_vector ( ( RS485_QUANTITY - 1 ) downto 0 ) := -- transmit output enable
           ( others => '0' );
-- quality informations (QI) 1=BAD 0=GOOD
LP_QI_BAD: in std_logic; -- quality information of L+ supply
DI_QI_BAD: in std_logic_vector ( ( DI_QUANTITY - 1 ) downto 0 ); -- quality information of digital process inputs (DIs)
DQ_QI_BAD: in std_logic_vector ( ( DQ_QUANTITY - 1 ) downto 0 ); -- quality information of digital process outputs (DQs)
RS485_QI_BAD: in std_logic_vector ( ( RS485_QUANTITY - 1 ) downto 0 ); -- quality information of RS485 channels
-- SIMATIC CPU STOP
CPU_STOP: in std_logic; -- SIMATIC CPU status 1=STOP 0=RUN
-- Isochronous Signals
T_DC: in std_logic; -- TDC time mark tic signal
T_I: in std_logic; -- TI time mark tic signal
T_I_LATCH: in std_logic; -- TI_LATCH time mark tic signal
T_O_LATCH: in std_logic; -- TO_LATCH time mark tic signal
T_O: in std_logic; -- TO time mark tic signal
);
```

end entity TFL_FAST_USER_e;

The following picture shows the schematic connections between the System-Logic and the User-Logic.

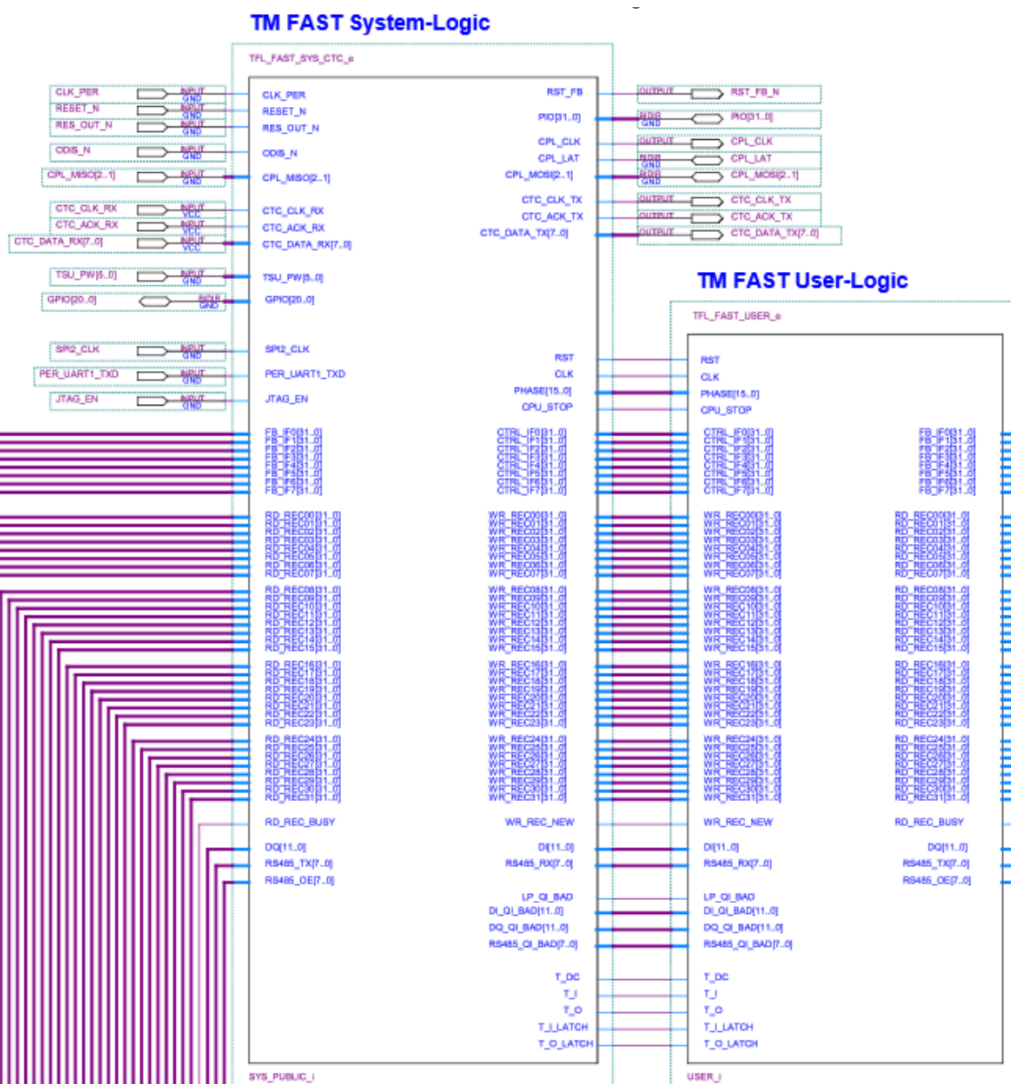


Figure 2: Logic design schematic top view (V2.0)

2.3 Structure of Product as Supplied

Designation	Description	MLFB	Unit pack ^{*)} per MLFB
TM FAST module	SIMATIC S7-1500, TM FAST Field Adaptable High-Speed Technology, programmable with Intel(R) Quartus(R) Prime software, 8xDI, 8xDQ, 4xDIQ 24 V, 8xRS485 DIQ, for extremely short response time of less than 1 µs	6ES7 554-1AA00-0AB0	1 ST
TM FAST debug connector	Connector to connect the module with a debugger over an Intel FPGA download cable	6ES7 554-1AA00-5AA0	1 ST
LTMFAST Library	TIA Library LTMFAST	-	SIOS download

*) Quantity unit (FDB) or base quantity unit (Key data in CADIM → SAP)

Possible values:

ST	Item	Standard packaging: 1 product in one package; price / 1 item Multipackaging: n items in one package; price / 1 item !!!!!
SZ	Set	Multipackaging: n products in one package; price / 1 set
PK	Pack	Multipackaging: n items in one package; price / 1 pack
M	Meter	Metered goods: n meters; price / 1m

3 Hardware Details

3.1 Hardware Introduction

The TM FAST module hardware is made of the 35mm width standard housing of the peripheral modules of the SIMATIC S7-1500 automation system and 3 PCBs:

- backplane PCB realizing the physical connection to the backplane bus (PBus+) of the SIMATIC S7-1500 automation system, hosting the backend ASIC (PER4-1 ASIC) and the Intel Cyclone10 LP FPGA, see [/BackplaneBoardSchematic/](#)
- One left function board containing the 24V power infeed, galvanic isolation, the CPL communication interface, and the process interface for 24V DI/DQs and RS485 IOs., see [/FunctionBoardSchematic/](#)
- One right function board galvanic isolation, the CPL communication interface, and the process interface for 24V DI/DQs and RS485 IOs, see [/FunctionBoardSchematic/](#)

3.2 Backplane Board Details

3.2.1 Backplane Block Diagram

Smart TM,CM - Backplane

Smart TM Cyclone 10 - A5E51601573
Smart TM Cyclone 3 - A5E51601574
CM Backplane - A5E51601575

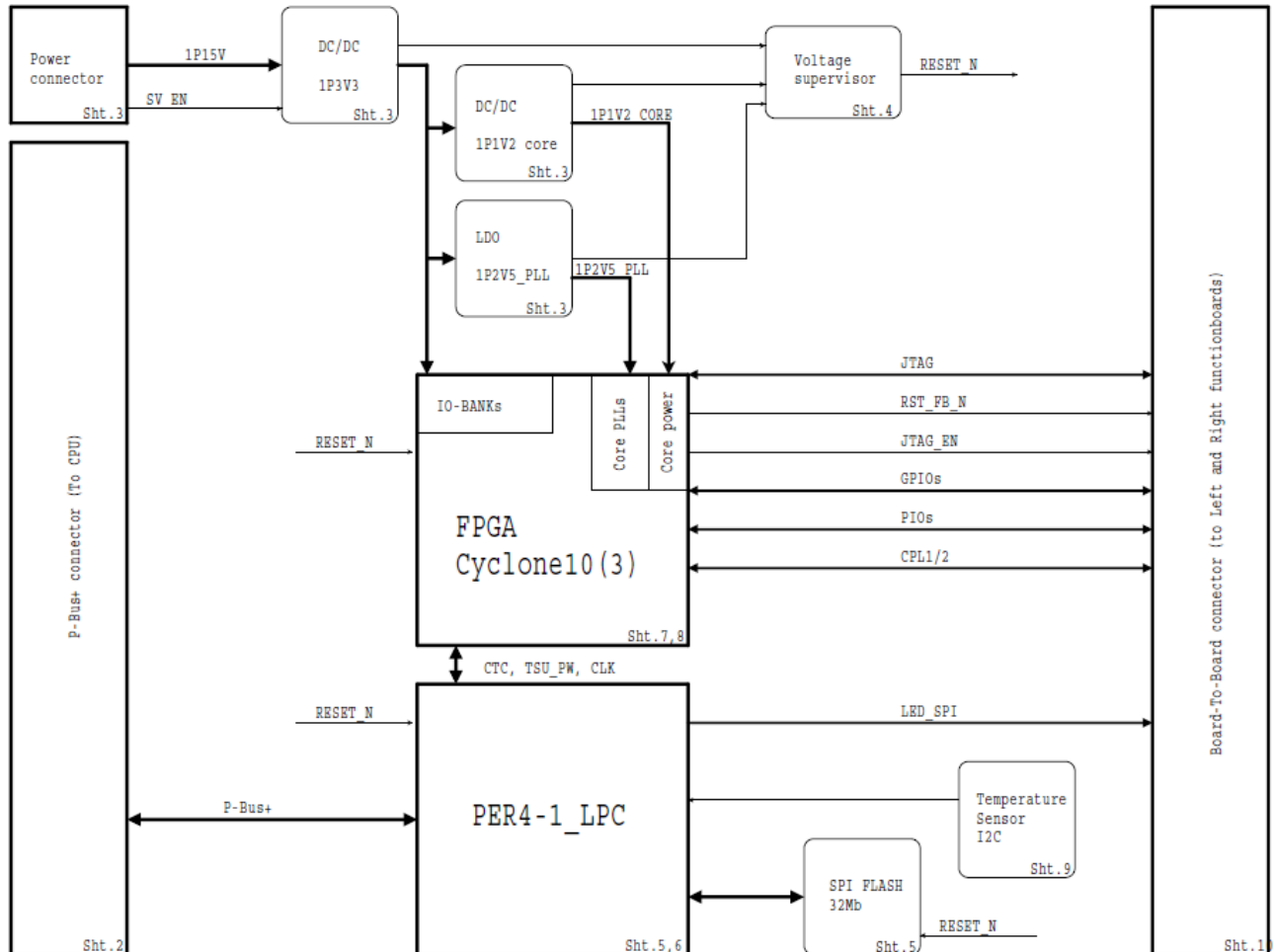


Figure 4: Backplane board diagram

3.2.2 CtC communication

The following configuration will be used to the CtC communication in both directions:

- ➔ Frequency 12,5MHz
- ➔ Number of lanes: 8
- ➔ Transfer acknowledgement: dedicated signal
- ➔ Transfer checksum: active
- ➔ Timeout: 3us (minimal possible value is 2,55us)
- ➔ No IRQ-Handling active
- ➔ CtC driving strength is set to 4mA

During CtC communication, if any error is detected, the FPGA configuration is removed from the FPGA and the diagnostic "Customer application not executable" is reported until the FPGA is loaded again.

3.2.3 Temperature measurement

TM FAST backend device implements an Inventra, MI2C - I2C bus interface and has a temperature sensor TMP100 assembled. TMP100 is an I2C compatible sensor with a measurement range -55 to 125 °C.

- Readable as a float value in the TMFASTControlREC
- Resolution: 0.25°C
- Current: 45 µA
- Accuracy:
 - ±1°C (Typical) from -55°C to 125°C
 - ±2°C (Maximum) from -55°C to 125°C
- I2C frequency: 100 kHz

For more details about I2C or TMP100, please refer to [/I2C and TMP100 sensor wiki/](#)

In the first versions of TM FAST (V1.0 and V2.0), the temperature will only be readable in the TMFASTControlREC, no internal monitoring will be implemented. In a later version, the temperature shall be monitored by the Backend firmware and trigger a diagnostic error "overtemperature" if the maximum acceptable value is reached.

The measured temperature value shall never get higher than 90°C according to the hardware temperature tests.

3.2.4 FPGA device

3.2.4.1 FPGA device type

The selected FPGA device for TM FAST is an Intel Cyclone 10 LP device, the precise ordering code is 10CL025YU256A7G, see the following picture for more details (selected device marked yellow).

Sample Ordering Code and Available Options for Intel Cyclone 10 LP Devices —Preliminary

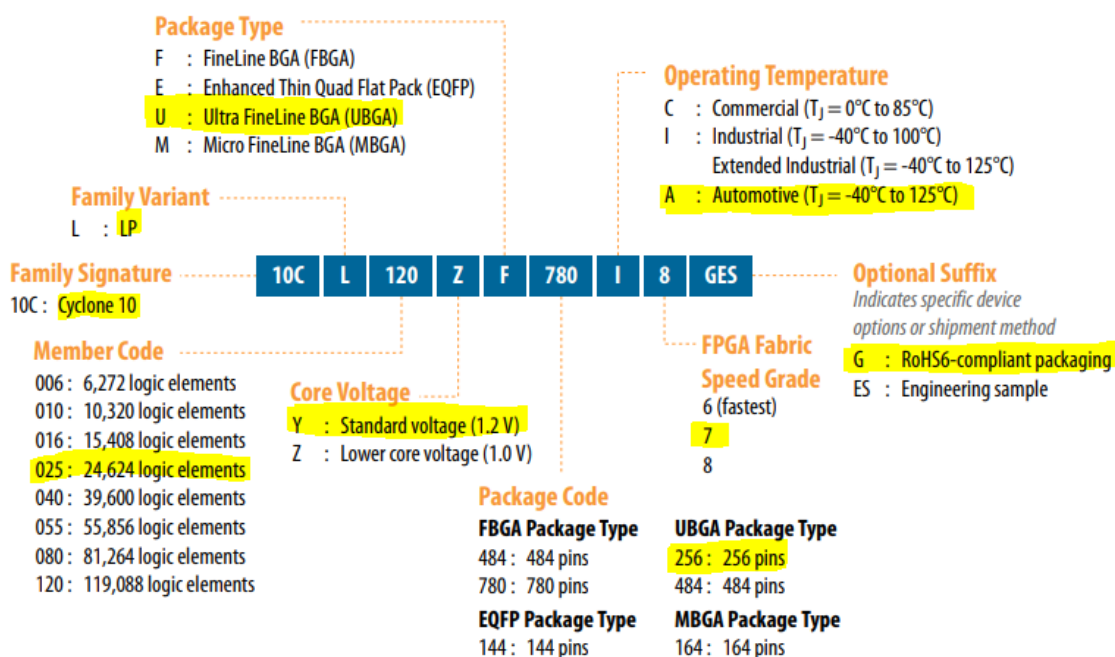


Figure 5: FPGA ordering code

Resource		Device							
		10CL006	10CL010	10CL016	10CL025	10CL040	10CL055	10CL080	10CL120
Logic Elements (LE)		6,272	10,320	15,408	24,624	39,600	55,856	81,264	119,088
M9K Memory	Block	30	46	56	66	126	260	305	432
	Capacity (Kb)	270	414	504	594	1,134	2,340	2,745	3,888
18 × 18 Multiplier		15	23	56	66	126	156	244	288
PLL		2	2	4	4	4	4	4	4
Clock		20	20	20	20	20	20	20	20
Maximum I/O		176	176	340	150	325	321	423	525
Maximum LVDS		65	65	137	52	124	132	178	230

Figure 6: Cyclone 10 LP Maximum Resources

The selected FPGA contains 24624 LUTs (look-up tables) integrated each one in one logic array bloc (LAB). One LAB consists in 16 logic elements (LE). An LE is the smallest unit of logic in the Intel Cyclone 10 LP device architecture. Each LE has four inputs, a four-input look-up table (LUT), a register, and output logic. The four-input LUT is a function generator that can implement any function with four variables.

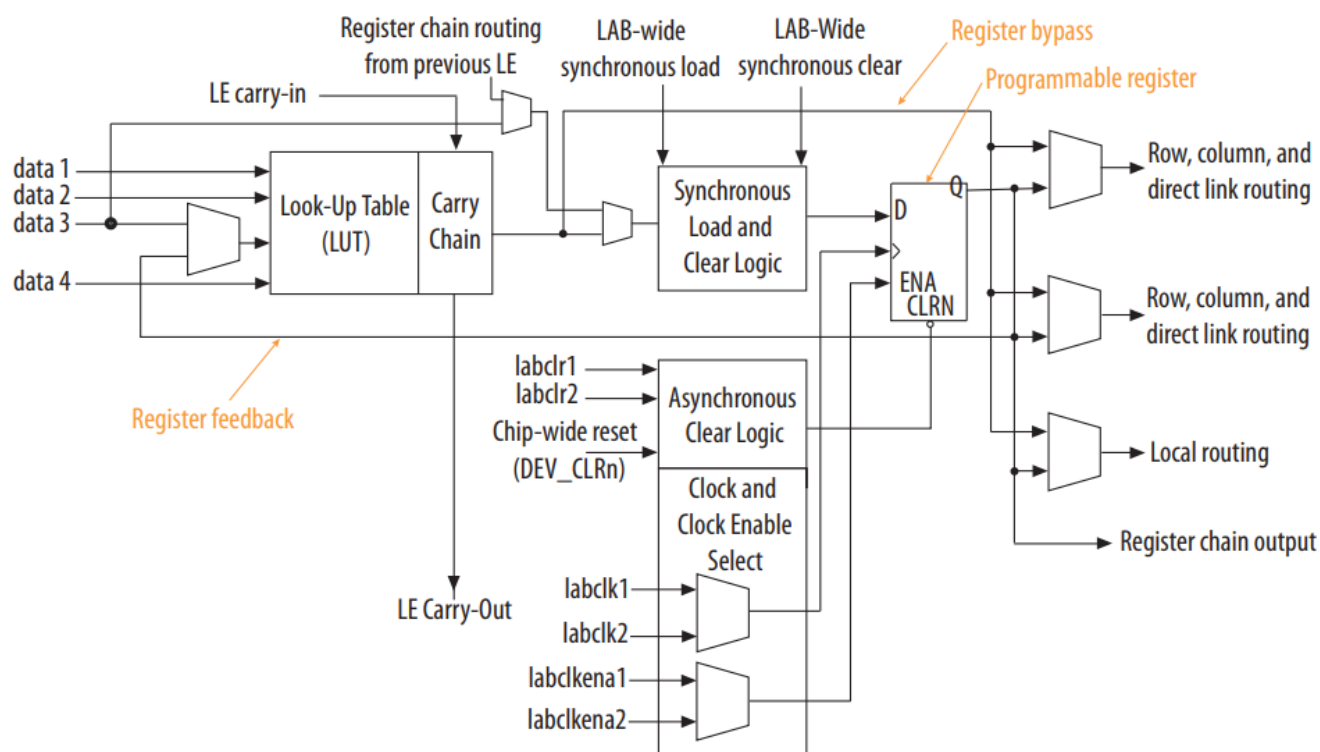


Figure 7: Intel Cyclone 10 LP Device Family LEs

For more details, refer to the Cyclone 10 LP Core Fabric and General Purpose I/Os Handbook: [/Cyclone10LPHandbook/](#)

3.2.4.2 Soft error detection within Cyclone 10 LP devices

The dedicated circuitry built in Intel Cyclone 10 LP devices consists of an error detection cyclic redundancy check (CRC) feature.

The circuitry principle detects the following:

- Configuration error
 - During FPGA configuration CRC is calculated within the FPGA and checked against in bit-stream incoming CRC
 - Configuration continues until the device detects an error or when all the values are calculated

- User Mode error detection
 - The contents of the CRAMs (configuration random-access memory) are monitored and soft errors are detected using 32-bit CRC computation
 - Monitoring is working continuously
 - This feature does not check memory blocks and I/O buffers
 - For testing purpose, it is possible to inject deliberately errors through the JTAG interface

Reaction of TM FAST to a soft error detection reported by the FPGA:

- ➔ The FPGA configuration is removed
- ➔ The diagnostic alarm “Customer application not executable” is reported to the CPU until the FPGA is successfully loaded again.

For more details about FPGA soft error detection, refer to the Cyclone 10 LP Core Fabric and General Purpose I/Os Handbook: [/Cyclone10LPHandbook/](#), Chapter “SEU Mitigation in Intel Cyclone 10 LP Devices”.

3.2.5 Flash device

The used flash device is a single SPI serial device with a 4Mbyte size. Its productive content is reed-solomon coded to detect and correct soft errors – only the FatalError block is not reed-solomon coded. TMFASTApplication will be stored in the FPGA blocks (FPGA 1 or FPGA 2) and their validity is stamped using dedicated readiness blocks (fpgaBlocksReadiness).

List of blocks within the flash device:

Flash content organization			
Block name	Start address	Size in bytes	End address
SBL_MP_Per4 1	0x00000000	20480 (20kB)	0x00005000
bootBlocksReadiness1.1	0x0000C000	4096 (4kB)	0x0000D000
bootBlocksReadiness1.2	0x0000D000	4096 (4kB)	0x0000E000
iumBlocksReadiness1.1	0x0000E000	4096 (4kB)	0x0000F000
iumBlocksReadiness1.2	0x0000F000	4096 (4kB)	0x00010000
SBL_MP_Per4 2	0x00010000	20480 (20kB)	0x00015000
bootBlocksReadiness2.1	0x0001C000	4096 (4kB)	0x0001D000
bootBlocksReadiness2.2	0x0001D000	4096 (4kB)	0x0001E000
iumBlocksReadiness2.1	0x0001E000	4096 (4kB)	0x0001F000
iumBlocksReadiness2.2	0x0001F000	4096 (4kB)	0x00020000
I&M block 1	0x00020000	12288 (12kB)	0x00023000
I&M block 2	0x00023000	12288 (12kB)	0x00026000
FW_Per4 1	0x00028000	495616 (484kB)	0x000A1000
FW_Per4 2	0x000A3000	495616 (484kB)	0x0011C000
HWInfo	0x00120000	20480 (20kB)	0x00125000
fpgaBlockReadiness1.1	0x00125000	4096 (4kB)	0x00126000
fpgaBlockReadiness1.2	0x00126000	4096 (4kB)	0x00127000
fpgaBlockReadiness2.1	0x00127000	4096 (4kB)	0x00128000
fpgaBlockReadiness2.2	0x00128000	4096 (4kB)	0x00129000
FPGA 1	0x00130000	790528 (772kB)	0x001F1000
FPGA 2	0x00200000	790528 (772kB)	0x002C1000
FatalError (Backend)	0x00300000	634880 (620kB)	0x0039B000
EMPTY	0x003FFFFFFF	0	0x003FFFFFFF

The flash device can be soldered empty during production and can be programmed using the manufacturing mode access. Note that the internal SBL (Secondary Bootloader) Boot-Mode of the PER4-1 ASIC is used during manufacturing mode access.



NOTE: Hint for testing

If an oversized upd file is generated for testing, the size of the firmware component should be bigger than 484kB, the size of the SBL component should be higher than 20kB.

3.3 Function boards details

3.3.1 Function boards block diagram

Function-Boards - TM FAST

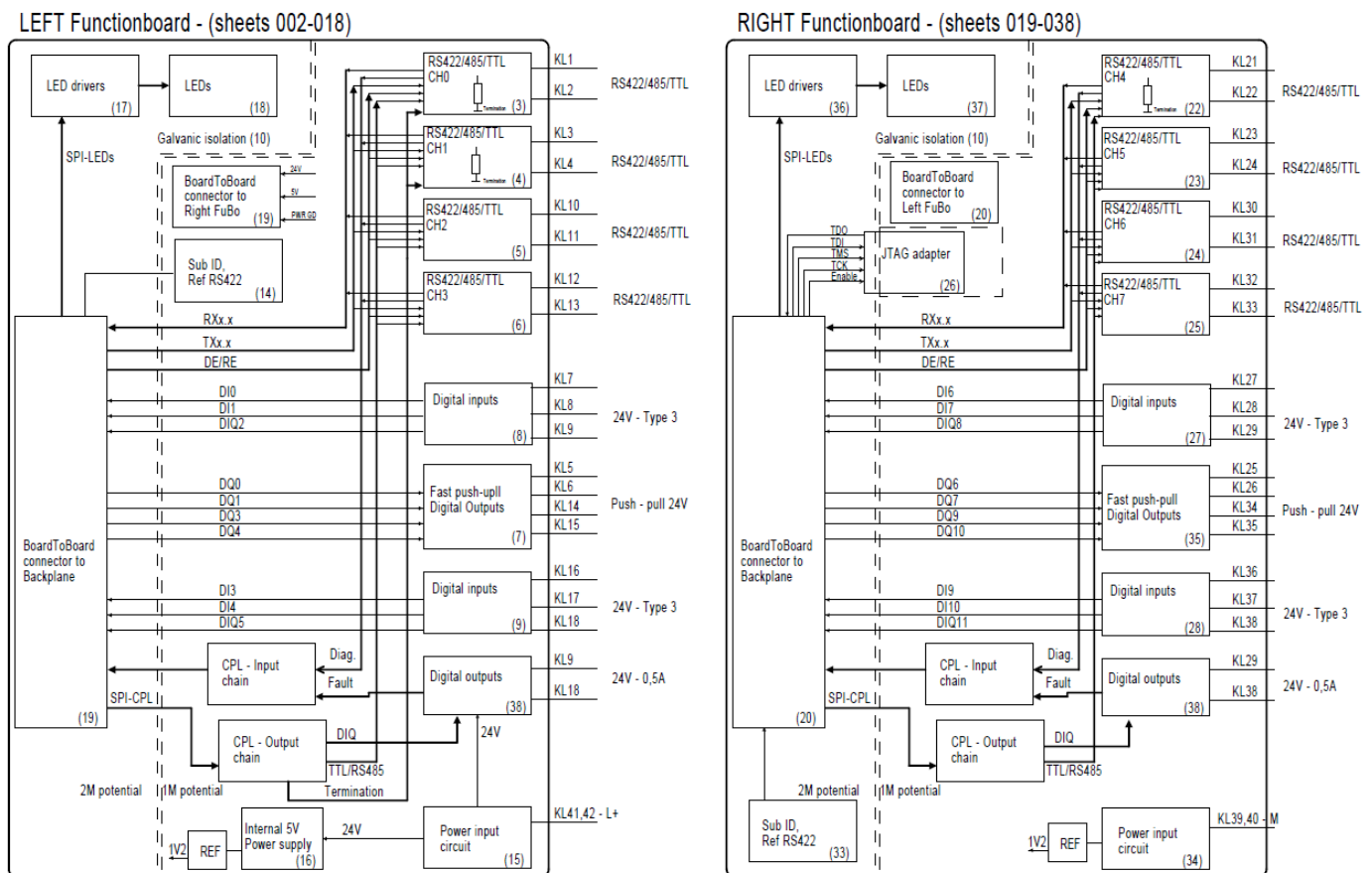


Figure 8: Function boards diagram

3.3.2 Hardware clamps functionalities

Left function board		Right function board	
CH0+/CH0-	RS485 – switchable direction (over GPIO) TTL - common parametrization for CH0&CH1 Switchable termination resistor (SSI, Incremental option) Dedicated wire-break detection	CH4+/CH4-	RS422 - switchable direction (over GPIO) TTL - common parametrization for CH4&CH5 Switchable termination resistor (SSI, Incremental option) Dedicated wire-break detection

CH1+/CH1-	RS422 - switchable direction (over GPIO) TTL - common parametrization for CH0&CH1 Switchable termination resistor (SSI, Incremental option) Dedicated wire-break detection	CH5+/CH5-	RS422 - switchable direction (over GPIO) TTL - common parametrization for CH4&CH5 no termination resistor Dedicated wire-break detection
DQ0	Push-pull driver, Frequency up to 200kHz Common overload detection (DQ0,1,3,4)	DQ6	Push-pull driver, Frequency up to 200kHz Common overload detection (DQ6,7,9,10)
DQ1	Push-pull driver, Frequency up to 200kHz Common overload detection (DQ0,1,3,4)	DQ7	Push-pull driver, Frequency up to 200kHz Common overload detection (DQ6,7,9,10)
DI0	24V digital input, Type 3, current sink 2mA	DI6	24V digital input, Type 3, current sink 2mA
DI1	24V digital input, Type 3, current sink 2mA	DI7	24V digital input, Type 3, current sink 2mA
DIQ2	24V digital input/output, if DQ used 10kHz, high-side driver with 0.5A Dedicated overload detection	DIQ8	24V digital input/output, if DQ used 10kHz, high-side driver with 0.5A Dedicated overload detection
CH2+/CH2-	RS422 - switchable direction (over GPIO) TTL - common parametrization for CH2&CH3 no termination resistors Dedicated wire-break detection	CH6+/CH6-	RS422 - switchable direction (over GPIO) TTL - common parametrization for CH6&CH7 no termination resistors Dedicated wire-break detection
CH3+/CH3-	RS422 - switchable direction (over GPIO) TTL - common parametrization for CH2&CH3 no termination resistors Dedicated wire-break detection	CH7+/CH7-	RS422 - switchable direction (over GPIO) TTL - common parametrization for CH6&CH7 no termination resistors Dedicated wire-break detection
DQ3	Push-pull driver, Frequency up to 200kHz Common overload detection (DQ0,1,3,4)	DQ9	Push-pull driver, Frequency up to 200kHz Common overload detection (DQ6,7,9,10)
DQ4	Push-pull driver, Frequency up to 200kHz Common overload detection (DQ0,1,3,4)	DQ10	Push-pull driver, Frequency up to 200kHz Common overload detection (DQ6,7,9,10)
DI3	24V digital input, Type 3, current sink 2mA	DI9	24V digital input, Type 3, current sink 2mA
DI4	24V digital input, Type 3, current sink 2mA	DI10	24V digital input, Type 3, current sink 2mA
DIQ5	24V digital input/output, if DQ used 10kHz, high-side driver with 0.5A Dedicated overload detection	DIQ11	24V digital input/output, if DQ used 10kHz, high-side driver with 0.5A Dedicated overload detection

Figure 9: Clamps functionalities

3.3.3 CPL communication

A serial communication is used between each function board and the system logic on the backplane board, the frequency used is 5MHz. The detailed content of the CPL input and output chains can be extracted from the schematic of the function boards.

The following filter values are setup in the system logic (file TFL_FAST_SYS_IP_CONF_MP_FAST_1_p.vhd):

- FuBo Detection Filter = 5,243 ms
- Basic CPL Inputs Filter = 1,311 ms
- DQ QI filter = 81,92 us
- DQ QI Monoflop Timer = 0,503 s

3.4 Clamps and LEDs description

Clamp	Name	Signal name (logic)	LEDs		LEDs		Signal name (logic)	Name	Clamp
			RUN	ERROR	MAINT				
1	CH0+	RS485(0)					RS485(4)	CH4+	21
2	CH0-		-			-		CH4-	22
3	CH1+	RS485(1)					RS485(5)	CH5+	23
4	CH1-		-			-		CH5-	24
5	DQ0	DQ(0)					DQ(6)	DQ6	25
6	DQ1	DQ(1)					DQ(7)	DQ7	26
7	DI0	DI(0)					DI(6)	DI6	27
8	DI1	DI(1)					DI(7)	DI7	28
9	DIQ2	DI/DQ(2)					DI/DQ(8)	DIQ8	29
10	CH2+	RS485(2)					RS485(6)	CH6+	30
11	CH2-		-			-		CH6-	31
12	CH3+	RS485(3)					RS485(7)	CH7+	32
13	CH3-		-			-		CH7-	33
14	DQ3	DQ(3)					DQ(9)	DQ9	34
15	DQ4	DQ(4)					DQ(10)	DQ10	35
16	DI3	DI(3)					DI(9)	DI9	36
17	DI4	DI(4)					DI(10)	DI10	37
18	DIQ5	DI/DQ(5)					DI/DQ(11)	DIQ11	38
19	P	-				-		M	39
20	-	-	-			-		M	40

Figure 10: Clamps and LED description

4 Acyclic communication

4.1 Module configuration

TM FAST is a S7-1500 module with the following configuration structure:

- MLFB: 6ES7 554-1AA00-0AB0
 - Module-ID: 0x6C20
 - number of channels: 1
 - number of subslots: 1
 - exactly one of the following submodules can be plugged in the subslot:
 - Submodule 0 (default) "FASTBig"
 - Submodule-ID: 0x00060102
 - Input data size: 32 Byte
 - Output data size: 32 Byte
 - Submodule 1 "FASTBig Profibus"
 - Submodule-ID: 0x00000102
 - Input data size: 32 Byte
 - Output data size: 32 Byte
 - Module-ID: 0x6C21
 - number of channels: 1
 - number of subslots: 1
 - only the following submodules can be plugged in the subslot:
 - Submodule 0 (default) "FASTBigIso"
 - Submodule-ID: 0x00060103
 - Input data size: 32 Byte
 - Output data size: 32 Byte
 - Submodule 1 "FASTBig Profibus"
 - Submodule-ID: 0x00000102
 - Input data size: 32 Byte
 - Output data size: 32 Byte
 - Submodule 2 "FASTBig" (not used by engineering)
 - Submodule-ID: 0x00060102
 - Input data size: 32 Byte
 - Output data size: 32 Byte



NOTE:

the Submodule 1 will only be used in case of TIA-Portal (mdd) use with a TM FAST behind a Profibus head station.

4.2 List of supported data records

The following records (in addition to standard ET 200MP records) are supported and detailed in the following chapters:

- TMFASTParaREC: parametrization record
- TMFASTUserWriteREC: customer specific write record
- TMFASTUserReadREC: customer specific read record
- TMFASTModuleInfoREC: module information record
- TMFASTControlREC: FPGA/application status and command record

4.3 Firmware update / Feature unlock record

The internal implementation of the firmware update protocol will be reused, the firmware component **FWUpdate** is instantiated – this implementation is the typical ET 200MP firmware update service implementation. For details about the firmware update protocol itself, refer to the corresponding [SPH/3/](#) or [FW-Update@SEA /2/](#).

Different than the V1.0 firmware, the V2.0 firmware supports secure firmware update which means a digital signature is part of the update file which confirms its authenticity. This digital signature is verified during the firmware update. If the validation fails, the customer will get an error message “The selected firmware file is not suitable for module xyz”. The signature for the V or P version is created by the trust-center (PPKI service). Only a few people have the authority to access that. More details are available in the [wiki](#). The versions for development have a different key pair. To be able to switch between these versions the feature unlock is required. The DS199 is used to unlock the module and exchange the public key on the device. If on the module there is a V version installed, it means the release public key is in use, only versions which are signed with the related private key can be installed. To install a development version, the user must first unlock the module, which requires the authentication and then the user might be able to set a new key for the signature validation. More details will be found in [FW-Update@SEA](#) Chap2.2. Integrity of FW/2/.

4.4 Module info record

The record TMFASTModuleInfoREC is implemented using:

- Record index 198
- Only available for development firmware (removed in release candidate and V-Version)
- Read and write record are possible
- A write record is required first to define which data shall be delivered for the next read accesses

4.4.1 TMFASTModuleInfoREC write

The write record length is 4 bytes as follow:

- Byte 0: Version
 - Only value 1 supported (ErrorCode 0xA8 reported if different)
- Byte 1: Command
 - 0: no command → not supported
 - 1: command isochronous measurements with reset → supported
 - 2: command hardware info → not supported
 - 3: command get acyclic information → not supported
 - 4: command isochronous measurements without reset → supported
 - 5: command get build time → supported
 - 6: command force ODIS → not supported
 - 0xFD: command trigger fatal error frontend → not supported
 - 0xFE: command deactivate watchdog → not supported
 - 0xFF: command trigger watchdog → not supported
- Byte 2: SubCommand1
 - Reserved (not used)
- Byte 3: Subcommand2

- Reserved (not used)

4.4.2 TMFASTModuleInfoREC read

By reading TMFASTModuleInfoREC, the same 4-byte header is returned as written with the last TMFASTModuleInfoREC write followed by the required data.

Supported data:

- Get build time – returns the date and time at which the firmware has been built
 - 4 bytes header (see write)
 - 20 bytes corresponding the following characters:
 - Date: “Mmm dd yyyy” (11 bytes) – example: “Jul 07 2022”
 - One space character “ ” (1 byte)
 - Time: “hh:mm:ss” (8 bytes) – example: “08:00:16”

4.5 Customer-specific data records

4.5.1 Customer-specific write record (TMFASTUserWriteREC)

The record TMFASTUserWriteREC is implemented using:

- Record index 101
- Record length defined by the customer in the file TFL_FAST_USER_IP_CONF_PUBLIC_p.vhd in the constant “WR_REC_DWORD_SIZE” – the length is fix for a specific logic version and cannot be changed in operation without changing the loaded logic.
- Possible length in bytes: 4,32,64 or 128
- Record can be written or read

Behavior in case no TMFASTApplication is active (read or write):

- ErrorCode1: 0xE1 and ErrorCode2: 0xA0

Behavior if the written record length is not equal to WR_REC_DWORD_SIZE:

- Writing a length (in bytes) different than WR_REC_DWORD_SIZE*4 will lead to the error code “write length error” - ErrorCode1: 0xB1 and ErrorCode2: 0x01.

4.5.2 Customer-specific read record (TMFASTUserReadREC)

The record TMFASTUserReadREC is implemented using:

- Record index 102
- Record length defined by the customer in the file TFL_FAST_USER_IP_CONF_PUBLIC_p.vhd in the constant “RD_REC_DWORD_SIZE” – the length is fix for a specific logic version and cannot be changed in operation without changing the loaded logic.
- Possible length in bytes: 4,32,64 or 128
- Read-only record

Behavior in case no TMFASTApplication is active:

- ErrorCode1: 0xE1 and ErrorCode2: 0xA0

4.6 Identification & Maintenance data

In addition to Identification & Maintenance 1-3, the TM FAST V2.0 supports the I&M4 data for security purpose (authentication of data at rest).

Here is the structure of the I&M4 payload data (extract from the SPH security):

IM_Signature (54 bytes)	Offset	Type	Name	Description	Include d		
	0	UINT32	USI	UserStructureIdentifier equals fixed value 0x63726331 (ASCII "crc1")			
	4	UINT32	chk_overall	overall checksum which combines all other checksums together (all local checksums and all subsidiary checksums)			
	8	UINT32	chk_overall_subs	overall checksum which combines all checksums from subsidiaries			
	12	UINT32	chk_static_local	checksum of the static data of this module			
	16	UINT32	chk_static_subs	combined checksum of all static data from subsidiary modules or NULL if there are no subsidiary modules			
	20	UINT32	chk_overall_setup	combined checksum of the setup data for the module and all its subsidiaries (if there are any)			
	24	UINT32	chk_remanent_local	checksum of the remanent data in this module			
	28	UINT32	chk_remanent_subs	combined checksum of remanent data from the subsidiary modules or NULL if there are no subsidiaries			
	32	UINT32	chk_working_local	checksum of the local working data			
	36	UINT32	chk_working_subs	combined checksum of working data from the subsidiary modules or NULL if there are no subsidiaries			
	40	byte[14]	unused	14 unused bytes (filled with zeros)			

The TM FAST will calculate/deliver the following values:

- chk_static_local: checksum over the I&M0 data
- chk_remanent_local: checksum over the I&M1-3 data
- chk_remanent_subs: checksum over the TM FAST user application loaded in the FPGA
- chk_working_local: checksum over the configuration and parametrization of the device
- chk_overall: checksum over all checksums
- chk_overall_subs: checksum over all checksums from subsidiaries
- chk_overall_setup: checksum over all remanent, working and user specific checksums



NOTE:

the checksum **chk_remanent_subs** will only reflect the signature of the application once an application has been loaded from the flash into the FPGA. If the application is not loaded or removed from the FPGA, the checksum is set back to the value **0xC704DD7B**.

- **I&M4 data is strictly read-only.**
The write access must be denied with PNIO-ErrorCode1 = Access/AccessDenied. The interface module must reject all write requests, acting on behalf of the entire device.
- **I&M4 must be readable.**
I&M-Supported bits at I&M0 must incorporate the I&M4 flag.
- **I&M4-Data should be generated upon request.**
The remanent storage is not necessary since I&M4 contains checksum/signature about the current parameters.

4.7 TMFASTApplication control

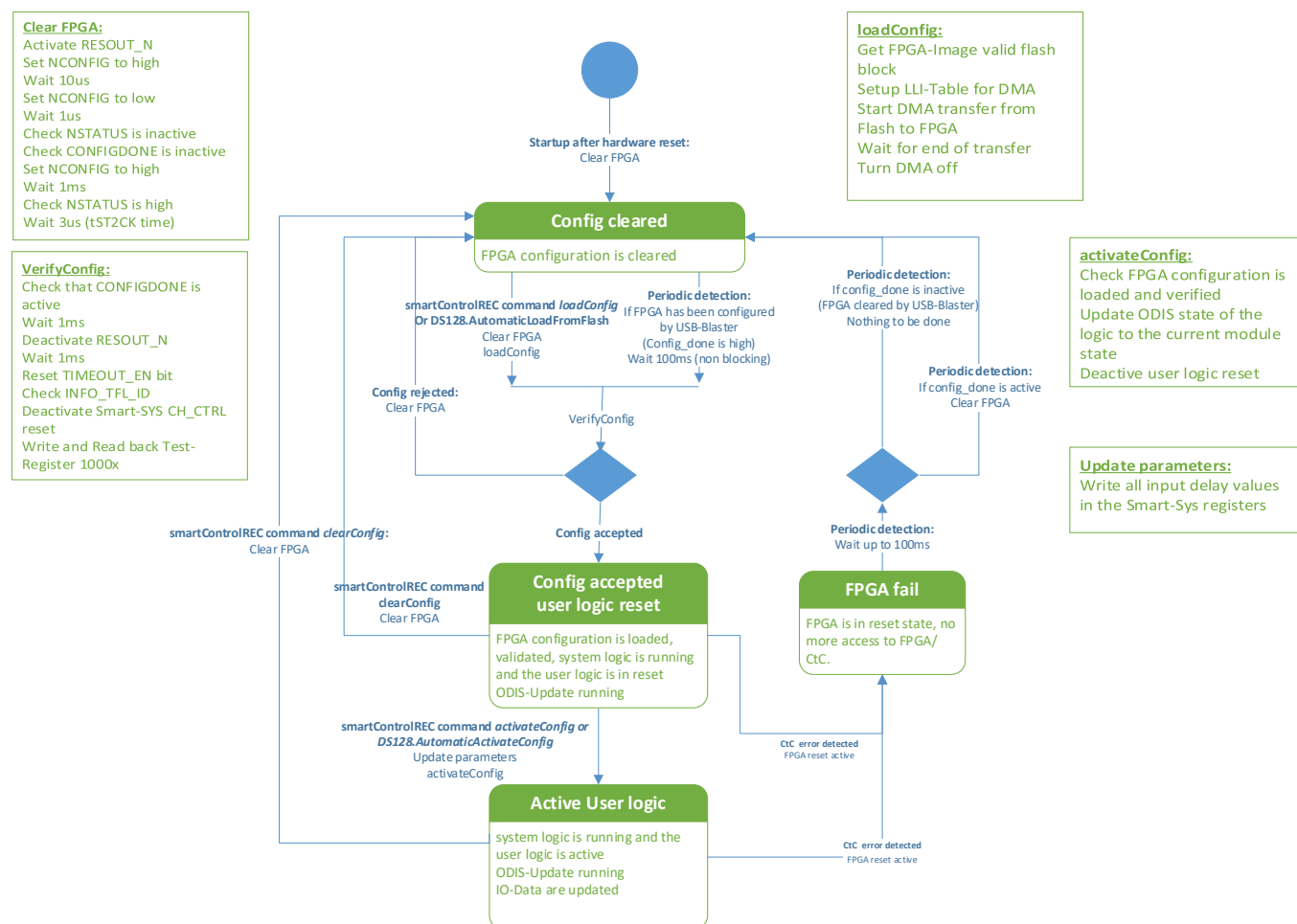
The state of the TM FAST module is defined by the state of TMFASTApplication.
The following states are possible for the TMFASTApplication:

- ➔ TMFASTApplication is cleared:
 - FPGA device is cleared and no communication between PER4-ASIC and FPGA is possible
- ➔ TMFASTApplication is loaded but not active:
 - FPGA device is loaded, the TFL_FAST_SYS logic can be reached by the PER4-ASIC firmware but the TFL_FAST_USER logic is kept in reset and do not get any clock
- ➔ TMFASTApplication is loaded and active:
 - FPGA device is loaded, the TFL_FAST_SYS logic can be reached by the PER4-ASIC firmware and the TFL_FAST_USER logic is running (reset inactive and clock active)

The state of the TM FAST module is controlled by the PER4-ASIC firmware, a change of state can be triggered by the user using the record TMFASTControlREC, using the USB-Blaster, using external tools to download the TMFASTApplication.

4.7.1 TMFASTApplication loading control

The following state machine is showing how the firmware is changing the state of the TM FAST module.



		16: Trigger an internal CtC-error (only for internal use, not accepted in V-version)	
		17: Trigger a CRC_ERROR of the FPGA (only for internal use, not accepted in V-version)	
		>17: reserved	

All commands between 1-4 requires ControlREC to be available (see bit "ControlREC is busy" in Read TMFASTControlREC) to be processed without error, otherwise the WRREC will be rejected – see list of errors below.

Details about the different commands:

- Load TMFASTApplication from flash:

This command will first clear the current application loaded on the FPGA if any present and then load a TMFASTApplication if any present in flash

- Activate a loaded TMFASTApplication

This command will activate a TMFASTApplication if any already application is already loaded in the FPGA. Activating means the reset of the user logic will be deactivated

- Clear TMFASTApplication

This command will clear the TMFASTApplication currently loaded on the FPGA if any present

- Delete TMFASTApplication from flash

This command will delete any TMFASTApplication from the flash device if any present

- Activate debug interface

This command will open the debug interface and therefore allow USB-Blaster debugger usage over the TM FAST debug connector. Note that the debug interface shall only be open during development phase and no more once the product is used in the the final industrial productive case.

- Deactivate debug interface

This command will close the debug interface to block access to the FPGA over the TM FAST debug connector.

Possible returned write record error codes:

If the length of the written record is wrong, the following error codes are reported:

- ➔ ErrorCode1: 0xB1 (WriteLengthError)
- ➔ ErrorCode2: 0x01

If no supported version number is used, the following error codes are reported:

- ➔ ErrorCode1: 0xD4 (VersionNotSupported)
- ➔ ErrorCode2: 0x00

If unsupported commands are executed, the following error codes are reported:

- ➔ ErrorCode1: 0xA9 (FeatureNotSupported)
- ➔ ErrorCode2: 0x00

If the FASTControlREC is busy while a command 1-4 is issued, the following error codes are reported:

- ➔ ErrorCode1: 0xA7 (moduleBusy)

➔ ErrorCode2: 0x00

4.7.2.2 Read TMFASTControlREC

ByteNr	Bit 7	Bit 6	Bit 5	Bit 4	Bit 3	Bit 2	Bit 1	Bit 0	DataType
0	Dataset version = 1								Byte
1	ModuleState								Byte
	Reserved			Control REC is busy	Debug interface state	FPGA state (see table below)			
2	Reserved								Byte
3	LastErrorCode (see table below)								Byte
4..7	SystemLogicVersion								DWord
8..11	UserLogicVersion								DWord
12..19	Application ID (8 characters)								Char[8]
20	Length of user write data record (bytes)								Byte
21	Length of user read data record (bytes)								Byte
22	Reserved								Byte
23	Reserved								Byte
RS485ChannelSettings									
24	Reserved CH0		Termination CH0		Mode CH0			Byte	
25	Reserved CH1		Termination CH1		Mode CH1			Byte	
26	Reserved CH2		Termination CH2		Mode CH2			Byte	
27	Reserved CH3		Termination CH3		Mode CH3			Byte	
28	Reserved CH4		Termination CH4		Mode CH4			Byte	
29	Reserved CH5		Termination CH5		Mode CH5			Byte	
30	Reserved CH6		Termination CH6		Mode CH6			Byte	
31	Reserved CH7		Termination CH7		Mode CH7			Byte	
32..35	Temperature								Real
36..39	Reserved (Function ID for licensing)								DWord
40..43	UserLogicVersion in Flash								DWord
44..51	Application ID (8 characters) in Flash								Char[8]
52..53	Timestamp TMFASTApplication in Flash - year								Word
54	Timestamp TMFASTApplication in Flash - month								Byte
55	Timestamp TMFASTApplication in Flash - day								Byte
56	Timestamp TMFASTApplication in Flash – weekday								Byte
57	Timestamp TMFASTApplication in Flash -hour								Byte
58	Timestamp TMFASTApplication in Flash -minute								Byte
59	Timestamp TMFASTApplication in Flash -second								Byte
60..63	Timestamp TMFASTApplication in Flash -nanosecond								DWord

4.7.2.2.1 Data record elements: FPGA state

These are the three application states of the FPGA controller in TM FAST.
The functionality of the user logic is only fully provided in state 2 “TMFASTApplication is loaded and active”.

FPGA state	
Value (dec)	Signification
0	TMFASTApplication is cleared
1	TMFASTApplication is loaded but not active
2	TMFASTApplication is loaded and active

4.7.2.2.2 Data record elements: Debug interface

This field is showing if the debug interface is activated or not. If the interface is activated (for TMFASTApplication development purpose only), accessing the FPGA by using the USB-Blaster and the TM FAST debug connector is possible. If the interface is deactivated (productive usage of the TM FAST module), the debug connector cannot be used.

Debug interface state	
Value (dec)	Signification
0	Debug interface is deactivated
1	Debug interface is activated

4.7.2.2.3 Data record elements: ControlREC is busy

This field is showing if the ControlREC internal state machine is busy or not. As long as it is busy, no new request using Write TMFASTControlREC (except commands debug activation/deactivation) can be processed and the value of the delivered state using Read TMFASTControlREC is not "final". It is recommended to poll the value until ControlREC is no more busy.

ControlREC is busy	
Value (dec)	Signification
0	ControlREC is available
1	ControlREC is busy

4.7.2.2.4 Data record elements: LastErrorCode

LastErrorCode during FPGA configuration checking		
Value (hex)	Signification	Text for customer
0	Success	Success
1	CTC error detected during communication, CRC failure	Failure communicating with TMFASTApplication, possible solution: update application
2	CTC error detected during communication, Timeout failure	Failure communicating with TMFASTApplication, possible solution: update application
3	CRC error detected during communication, Bus failure	Failure communicating with TMFASTApplication, possible solution: update application
4	CTC error detected during communication, Bus AHB Master failure	Failure communicating with TMFASTApplication, possible solution: update application
5	no CONF_DONE signal active (unloaded)	Failure communicating with TMFASTApplication, possible solution: update application
6	Test pattern check failure	Failure communicating with TMFASTApplication, possible solution: update application
7	FPGA is not loaded and cannot be activated	TMFASTApplication cannot be activated, load application first
8	FPGA clear error	TMFASTApplication cannot be cleared from FPGA, possible hardware defect
9	No TMFASTApplication available in flash	No TMFASTApplication available in flash, update application in flash
A	Wrong ID in TMFASTApplication detected	Failure communicating with TMFASTApplication, possible solution: update application
B	FPGA CRC error detected	Failure communicating with TMFASTApplication, possible solution: update application
C	TMFASTApplication License not available	License for TMFASTApplication not available

4.7.2.2.5 Data record elements: SystemLogicVersion / UserLogicVersion

Provide information about the currently loaded System- and UserLogicVersion.
The logic versions are built up as following pattern: **LetterMajor.Minor.Patch**
E.g., **R1.1.32** – digits are decimal digits (0 to 99).

SystemLogicVersion / UserLogicVersion		
ByteNr	Signification	Data Type
0	Major	Byte
1	Minor	Byte
2	Patch	Byte
3	Letter	Char

4.7.2.2.6 Data record elements: Termination CHx

This field describes the termination type of the corresponding RS485 IO, useful in case of bus implementation. The value reported in this field corresponds to the settings RS485_TERMINATION in the user configuration (see **Error! Reference source not found.**).

Termination CHx (RS485 or RS422)	
Value (hex)	Signification
0	Termination is off for channel CHx
1	Termination is on for channel CHx
2	Termination state is defined by the parametrization: see field "Bus termination" in TMFASTParaREC
3	Termination not available on channel CHx

4.7.2.2.7 Data record elements: Mode CHx

Mode CHx (RS485, RS422 or TTL)	
Value (hex)	Signification
0	CHx is used as RS485 bidirectionnal
1	CHx is used as RS422 input
2	CHx is used as RS422 output
3	CHx is used as TTL input
4	CHx is used as TTL output
5..14	Reserved
15	CHx is not used

4.7.3 TMFASTApplication download

The selected format for transferring the TMFASTApplication download file is the standard "upd" file format as defined in the firmware-update SPH. The advantage of this format is the possibility to use any standard SIMATIC firmware-update tool to download the TMFASTApplication, for example TIA-Portal, Webserver, Automation Tool, MFCT, ...

This update file is generated from the FPGA programming file (RBF file) using the MFCT tool. The update file contains the information which data record index is used (index of TMFASTAppDownloadREC is used) and the index will be recognized by the backend firmware to differentiate between a firmware update and a TMFASTApplication update. Once the TMFASTApplication download is successfully performed, the TMFASTApplication will be available non-volatile on the flash device on the module.

4.7.3.1 Details about the application download update file

The update file for the TMFASTApplication download has the following specificities:

- Record index used is 238
- Update file contains the HEADER, BG_ABL and one additional component specific for the one customer TMFASTApplication
- Component name generated by MFCT is CL_001, component name generated by the firmware building environment is FPGAIM. The TM FAST firmware will accept both namings and using a different name will allow to make differentiations in the future if required
- The TMFASTApplication component contains the following information:

- Information about the FPGA programming file stored in one reed-solomon block and in the following order:
 - Version of this FPGA information structure → 1 Byte (Value: 1)
 - Date and time of the upd generation → 12 Bytes: Year (2 Bytes), Month (1 Byte), Day (1 Byte), Weekday (1Byte), Hour (1 Byte), Minute (1 Byte), Second (1 Byte), Nanosecond (4 Bytes): note that the data type used here is the same than “DTL” in SCL language (see TIA online help for more information).

DTL

Byte	Component	Data type	Value range
0	Year	UINT	1970 to 2262
1			
2	Month	USINT	1 to 12
3	Day	USINT	1 to 31
4	Weekday	USINT	1 (Sunday) to 7 (Saturday) The weekday is not considered in the value entry.
5	Hour	USINT	0 to 23
6	Minute	USINT	0 to 59
7	Second	USINT	0 to 59
8	Nanosecond	UDINT	0 to 999999999
9			
10			
11			

- Net length in bytes of the FPGA programming file before reed solomon encoding → 4 Bytes
- TMFASTApplication ID → 8 Bytes
- TM FAST user-logic version → 4 Bytes

Byte	1	2	3	4	5	6	7	8
0	Structure version = 1							
1	Year							
2	Month							
3	Day							
4	Weekday							
5	Hour							
6	Minute							
7	Second							
8	Nanosecond							
9								
10								
11								
12	Net length of RBF file							
13								
14								
15								
16	AppID							
17								
18								
19								
20								
21								
22								
23								
24								

25	Version (same version as before, can be ignored)
26	
27	
28	
29..235	Reserved (0x00)
236..252	Reed-Solomon checkbytes

- FPGA programming file Reed-Solomon coded

4.7.3.2 *TMFASTApplication download record (TMFASTAppDownloadREC)*

A specific record (TMFASTAppDownloadREC) is used to download the TMFASTApplication into the flash device of the module. This record is implementing the same protocol than firmware-update as described in the SPH "SPH AS Mechanisms, Volume 3: E1 Firmware Update".

The index of the record TMFASTAppDownloadREC is 238 (usually index 239 is used for firmware update) and will be supported by all traditional firmware-update tools (TIA-Portal, Automation Tool, MFCT-Tool, CPU-Webserver, ...).

At the end of the TMFASTApplication download process, a "reset request" (see checkbox "Run firmware after update" in TIA-Portal) can be sent to the module and the behavior in this case is depending on the module parametrization (see TMFASTParaREC)

Case "Load TMFASTApplication at startup" is active

- ➔ At reception of "reset request", the newly downloaded TMFASTApplication is loaded into the FPGA
- ➔ If "Always activate TMFASTApplication" is active, the newly loaded application is activated

Case "Load TMFASTApplication at startup" is not active

- ➔ No action is triggered by the "reset request" command and a running TMFASTApplication will remain unaffected.

Note that if an update (firmware or application update) is already running, another one will be rejected until the running one is finished (or cancelled).

4.7.3.3 *TMFASTApplication download using Firmware-Update tool*

Any Siemens firmware update tool shall be used to download the application: Siemens Automation Tool, MFCT, Webserver, CPU memory card or TIA Portal firmware update.

4.7.4 Compatibility Firmware / System logic

For information regarding the compatibility between firmware and system logic, please refer to chapter 7

4.8 Module parametrization

4.8.1 Parametrization record (TMFASTParaREC)

4.8.1.1 *Parametrization record description*

The data record "TMFASTParaREC" (record index 128) is written to the TM FAST module to setup the behavior of the frontend device. Some parameters are dedicated to the TFL_FAST_SYS logic and must be written by the backend firmware in the frontend device each time a new TMFASTApplication is activated in the FPGA.

The following registers of TFL_FAST_SYS are updated prior to TMFASTApplication activation:

- ➔ DI filter time values
- ➔ Bus termination (RS485 CH0, CH1 and CH4 only)
- ➔ RX (RS485 input) filter time values
- ➔ Diagnostic masks to activate the red LEDs in case of active diagnostics

The TMFASTParaREC record can be used to change the parametrization of TM FAST during CPU runtime.

It has a static length of 32 bytes, where the first 4 bytes contain header information and the following 28 bytes channel parameters.

4.8.1.2 Parametrization record structure

The following record structure is valid for the V2.0 firmware. The difference between the record for V1.0 and V2.0 is the record version ("1" for V1.0 and "2" for V2.0) and the 2 new V2.0 additional parameters UserProcessingTimeInput and UserProcessingTimeOutput.

ByteNr	Bit 7	Bit 6	Bit 5	Bit 4	Bit 3	Bit 2	Bit 1	Bit 0
Header								
0	reserved		major version = 0		minor version = 2 ¹			
1	channel parameter length = 28							
2	reserved							
3	reserved							
ByteNr								
Base-Parameters								
4	reserved		Behavior at CPU-STOP	Load TMFASTApplication at startup	Always activate TMFASTApplication	Wire-break diagnostic interrupt	Digital output error diagnostic interrupt	Load-voltage diagnostic interrupt
DI0-Parameters								
5	Reserved				DI-Input delay			
DI1-Parameters								
6	Reserved				DI-Input delay			
DIQ2-Parameters								
7	Reserved				DI-Input delay			
DI3-Parameters								
8	Reserved				DI-Input delay			
DI4-Parameters								
9	Reserved				DI-Input delay			
DIQ5-Parameters								
10	Reserved				DI-Input delay			
DI6-Parameters								
11	Reserved				DI-Input delay			
DI7-Parameters								
12	Reserved				DI-Input delay			
DIQ8-Parameters								
13	Reserved				DI-Input delay			
DI9-Parameters								
14	Reserved				DI-Input delay			
DI10-Parameters								
15	Reserved				DI-Input delay			
DIQ11-Parameters								

¹ Only supported by firmware V2.0

16	Reserved		DI-Input delay
CH0..7 - Parameters			
17..24	Reserved	Bus termination CHx	RX-Filter time
Reserved Bytes			
25..27	reserved		
Isochronous user application processing times			
28..29	UserProcessingTimeInput ² (0...4000us)		
30..31	UserProcessingTimeOutput ³ (0...4000us)		

4.8.1.2.1 Data record elements: DI-Input delay

Input delay			Shielding/Noise filtering
Value (dec)	Signification	Number of ticks (75MHz)	
0	0.001ms	74	customer specific (extra HW-shielding/filtering and noise filter in User Logic)
1	0.003ms	222	shielded wires (Siemens standard setup)
2	0.05ms	3712	shielded wires (Siemens standard setup)
3	0.1ms (default)	7425	unshielded wires
4	0.4ms	29700	unshielded wires
5	0.8ms	59400	unshielded wires
6	1.6ms	118800	unshielded wires
7	3.2ms	237600	unshielded wires
8	12.8ms	950400	unshielded wires
9	20ms	1484999	unshielded wires
10..14	reserved	-	
15	customer constant	defined by logic	

In case "customer constant" is selected for a specific digital input, the active filter time is coming from the VHDL package file "TFL_FAST_USER_IP_CONF_PUBLIC_p.vhd" in the constant "DI_FILTER_USER_VAL_MS".

4.8.1.2.2 Data record elements: RX-Filter time

RX-Filter time		
Value (dec)	Signification	Number of ticks (75MHz)
0	0.1us	7
1	0.2us	14
2	0.5us	37
3	1us	74
4	2us	148
5	5us	371
6	10us	742
7	20us	1485
8	50us	3712
9	100us (default)	7425
10	200us	14850
11	500us	37125
12	1000us	74250
13, 14	reserved	-

² Only supported by firmware V2.0, in freewheeling mode will be ignored and 10us will be used (V2.0 only)

³ Only supported by firmware V2.0, in freewheeling mode will be ignored and 10us will be used (V2.0 only)

15	customer constant	defined by logic
----	-------------------	------------------

In case “customer constant” is selected for a specific digital input, the active filter time is defined in the VHDL package file “TFL_FAST_USER_IP_CONF_PUBLIC_p.vhd” in the constant “RS485_FILTER_USER_VAL_MS”.

4.8.1.2.3 Data record elements: Bus termination CHx

The bus termination parameter is only considered if the TMFASTApplication has activated on the channel CHx termination “Termination state is defined by the parametrization” and if the CHx is supporting any termination (only supported by CH0, CH1 and CH4).

Bus termination CHx	
Value (dec)	Signification
0	Bus termination on CHx is off if parametrizable (default)
1	Bus termination on CHx is on if parametrizable

4.8.1.2.4 Data record elements: Load TMFASTApplication at startup

Load TMFASTApplication at startup	
Value (dec)	Signification
0	TMFASTApplication is loaded using TMFASTControlREC in the user program or USB-Blaster
1	TMFASTApplication is loaded at startup (default)

Additional information for value “TMFASTApplication is loaded at startup”: in the following scenarios the TMFASTApplication will be loaded:

- ➔ at reception of TMFASTParaREC:
 - no TMFASTApplication is already loaded in the FPGA (active or not)
 - TMFASTApplication is available in flash
- ➔ after downloading a new TMFASTApplication over TMFASTAppDownloadREC:
 - at reception of reset request (firmware update protocol) – see checkbox “Run firmware after update” in TIA Portal
- ➔ after downloading a new TM FAST firmware:
 - at reception of reset request (whole module is reset in this case) – see checkbox “Run firmware after update” in TIA Portal [TEST: will be tested manually or once the FB firmware-update is implemented]

4.8.1.2.5 Data record elements: Diagnostic interrupt

One bit is dedicated to one possible diagnostic source: wire-break (for RS485 inputs), digital output error (for 24V outputs) or load voltage supply. Each diagnostic can be activated/deactivated separately.

Wire-break/Digital output error/ load voltage Diagnostic interrupt	
Value (dec)	Signification
0	Corresponding module specific diagnostic interrupt is disabled (default)
1	Corresponding module specific diagnostic interrupts is enabled

4.8.1.2.6 Data record elements: Behavior at CPU-STOP

Behavior at CPU-STOP	
Value (dec)	Signification
0	Disable all outputs at CPU-STOP (default)

1	Behavior implemented in the TMFASTApplication
---	---

4.8.1.2.7 Data record elements: user application isochronous processing times

The following parameters are only supported by the firmware V2.0 if the isochronous mode is active. If freewheeling is active, the default value of 10us will be used internally, ignoring the value set in this parameter.

User application input/output processing time (us)	
Value (dec)	Signification
10	10us user application processing time (default)
0..4000	Input/output processing time value in 1us units

4.8.2 Parameter validation

Version check (Module ID V1.0 is projected):

If the version of the record is not exactly 1 (see header of the TMFASTParaREC, fields major and minor version), the following error code is returned:

- ➔ ErrorCode1: 0xE0 (InvalidStructure)
- ➔ ErrorCode2: 0x01

Version check (Module ID V2.0 is projected):

If the version of the record is not 1 or 2 (see header of the TMFASTParaREC, fields major and minor version), the following error code is returned:

- ➔ ErrorCode1: 0xE0 (InvalidStructure)
- ➔ ErrorCode2: 0x01

Record length check:

If the total record length is not exactly 32 bytes (channel length added to header length), the following error code is returned:

- ➔ ErrorCode1: 0xB1 (WriteLengthError)
- ➔ ErrorCode2: 0x01

Channel length check (only visible if record length check is successful):

If the channel length of the record is not exactly 28 bytes (see header of the TMFASTParaREC, field channel parameter length), the following error code is returned:

- ➔ ErrorCode1: 0xE0 (InvalidStructure)
- ➔ ErrorCode2: 0x02

Reserved fields set to 0:

If any reserved field is set to any other value than "0", the following error code is returned:

- ➔ ErrorCode1: 0xE1 (Module specific error code)
- ➔ ErrorCode2: 0xF0

DI Input-delay range violation:

If an unallowed value is set as DI input-delay (reserved range 10..14), the following error code is returned:

- ➔ ErrorCode1: 0xE1 (Module specific error code)
- ➔ ErrorCode2: 0x01

RX Filter-time range violation:

If an unallowed value is set as DI input-delay (reserved range 13..14), the following error code is returned:

- ➔ Errorcode1: 0xE1 (Module specific error code)
- ➔ Errorcode2: 0x02

Bus termination CHx:

If a bus termination is turned on (value "1") on CHx and CHx does not support any termination (only supported by CH0, CH1 and CH4)

- ➔ Errorcode1: 0xE1 (Module specific error code)
- ➔ Errorcode2: 0x03

Isochronous timings validation (Firmware >= 2.0):

First, the value of the user application processing time values is to be checked:

If the parametrized value is greater than 4000us, the following error code will be reported:

- ➔ Errorcode1: 0xE1 (Module specific error code)
- ➔ Errorcode2: 0x04

The isochronous timings CTI (ClockSyncCTIModule) and CTO (ClockSyncCTOModule) are depending on the new parameter values UserInputProcessingTime and UserOutputProcessingTime and therefore one parametrization shall not be accepted if the TI (TimelOInput) or TO (TimelOOutput) are not sufficient to run the required timing. CTI and CTO are calculated and checked by the MDD. This validation error will only come if the parametrization record is sent manually by the customer.

20	TimelOInput	U32	0 ... 4 000 000	Time base is ns	$\geq \frac{T_IO_InputMin}{ClockSyncCTIModule}$ & $\leq 4\,000\,00$
24	TimelOOutput	U32	0 ... 4 000 000	Time base is ns	$\geq \frac{T_IO_OutputMin}{ClockSyncCTOModule}$ & $\leq 4\,000\,00$

- ➔ Errorcode1: 0xE1 (Module specific error code)
- ➔ Errorcode2: 0xFF

4.8.3 Validation of clock-synchronised parameters

- ➔ At reception of the parametrization record:
 - UserProcessingTimeInput must fit in TimelOInput (see exact formula in Table 2: Isochronous timing table V2.0
 -)
 - UserProcessingTimeOutput must fit in TimelOOutput (see exact formula in Table 2: Isochronous timing table V2.0
 -)

4.8.4 Default parametrization

The default parametrization is depicted in **bold** in the chapter 4.8.1.2.

4.9 Service data

The service data from the device can be fetched using the standard TIA online&diagnosis function "Read service data". For more details about how to interpret the service data, refer to the following document: [link](#)

The service data contain the following information:

- ➔ Backend firmware trace: current trace from the PER4 RAM memory
- ➔ Backend firmware fatal error trace: trace from the last fatal error stored in the flash device
- ➔ No frontend information and no module information are available (different than for T-ASIC modules)

4.10 List of supported diagnostics interrupts

Diagnostic alarm	Error code	Scope	De-/activable?	Meaning	Corrective measures
Load voltage missing	11 _H	Channel	Yes	- Missing or insufficient supply voltage L+ - Wiring of supply voltage L+ faulty	- Check wiring of supply voltage L+ - Check total consumption of the load group
Error at digital outputs	10F _H	Channel	Yes	Problem at digital outputs of module Possible causes: - Short-circuit / Overload / external infeed	- Check wiring at the digital outputs - Check consumers connected to the digital outputs
RS422/TTL Fehler	502 _H	Channel	Yes	Error at the RS422/TTL input Possible cause: wire-break	Check wiring
Parameter error	10 _H	Module	No	The received parameter data record is invalid	Check parameter data record
Overtemperature	506_H	Module	No	Overtemperature detected by PER4 from the temperature sensor TMP100 device	
Customer application not executable	10B9 _H	Channel	No	A communication error has been detected with the FPGA and the application has been removed	Check that no timing violations have been reported during the FPGA synthesis and update the application of the module Other possible cause: FPGA is reporting a CRC error or no license is available for this application
Debug interface activated (Maintenance diagnostic)	10B8 _H	Channel	No	The debug interface is designated for commissioning and service purposes only.	Deactivate the debug interface using the corresponding TMFASTControlREC command and remove the TM FAST debug connector to ensure a correct behavior of the module at higher temperatures.

Table 1 Diagnostic interrupts of a TM FAST submodule

Note that both diagnostics “Issue detected with customer application” and “Debug-Interface is activated” are new and the final text will only be available in TIA V18.

Note that the diagnostic “Debug-interface is activated” is reported by the TM FAST module combined with a maintenance request. This means the module will not blink red but will show the yellow maintenance LED until the debug-interface is deactivated. The diagnostic message is visible in the CPU just like another diagnostic in the diagnostic buffer.

The diagnostics “Error at digital outputs” and “RS422/TTL error” are only reported by the module if the TMFAST application is loaded and active.

The diagnostic interrupt “Overtemperature” is not planned for the first revisions (V1.0, V2.0) of the module but could be activated in a later version.

4.11 TMFASTApplication licensing concept

The licensing concept is not planned for the first revisions of the TM FAST (V1.0 and V2.0). The idea of the licensing is to allow a TM FAST application developer to protect and limit the usage of the developed application by requiring a license. The License would be available in the S7-1500 CPU, the TM FAST module will check the availability of one license before activating a protected TMFASTApplication. One application is protected if the parameter "Function ID" is defined in the application with a value different than 0.

For security reason the V1.0 and V2.0 releases of the TM FAST module will reject the activation of the application if the "Function ID" defined in the application is not "0" and the diagnostic "Customer application not executable" will be reported to the CPU in case an activation is requested. The diagnostic is reported until the FPGA is loaded again.

5 Cyclic communication

5.1 Output data

The implemented submodule supports 32 bytes output data organized as an array of 8 double words (one double word consists of 32 bits).

5.2 Input data

The implemented submodule supports 32 bytes input data organized as an array of 8 double words (one double word consists of 32 bits).

5.3 Behaviour on ODIS (Output disable)

The information ODIS (Output Disable) is active in the module in case

- the CPU is turned to the STOP state
- or
- if the connection to the CPU is lost (fieldbus connection is lost)

In this particular scenario, the output data are no more updated by the CPU, only the input data coming from the module are still updated. The user logic gets the information about the ODIS state over the input signal CPU_STOP and can decide what should happen within the frontend device.

TM FAST firmware behavior in case of active ODIS:

- TMFASTApplication is loaded and active
 - output data are not updated into the FPGA user-logic
 - input data are copied from the FPGA user-logic to the backplane bus
 - behavior of the outputs is depending on the parameter "Behavior at CPU-STOP":
 - if set to "Disable all outputs at CPU-STOP", all outputs are disabled overruling any specific implementation in the TMFASTApplication
 - if set to "Behavior implemented in the TMFASTApplication", the behavior of the outputs is defined within the TMFASTApplication
- TMFASTApplication is not loaded at all or is loaded and not activated
 - output data are ignored
 - input data are all set to "0"
 - all outputs are disabled (no application loaded)

5.4 Isochronous mode

The TM FAST submodule supports both freewheeling and isochronous mode.

5.4.1 Isochronous features

Isochronous mode for Profibus is not supported.

Following features will be supported by the module:

- The TM FAST submodule is a T_i/T_o -module.
- The user logic gets as inputs the T_i , T_o and T_{DC} pulses → length of pulses: 1us
- The V2.0 firmware supports the parametrization of user specific processing times for inputs and outputs in the user application (see parameters UserProcessingTimeInput and UserProcessingTimeOutput)

The following wires are used between PER4 and FPGA for transmitting the isochronous pulses:

- TSU_PW0: T_{DC}
- TSU_PW2: T_O
- TSU_PW3: T_O_LATCH (only for Firmware V2.0)
- TSU_PW4: T_I_LATCH (only for Firmware V2.0)
- TSU_PW5: T_I

Used comparators in the PER4 ASIC (MSSYNCTimer):

Comparator 0: TDC	→ required for TSU_PW0
Comparator 1: TO with offset (TO0_PRE)	→ required for TIMER1 and TIMER2 settings
Comparator 2: TO without offset	→ required for TSU_PW2
Comparator 3: TO with offset (TO0_PRE)	→ required for TSU_PW3
Comparator 4: TI with offset	→ required for TSU_PW4
Comparator 5: TI without offset	→ required for TSU_PW5 and for IRQ trigger in the firmware

5.4.2 Isochronous implementation description

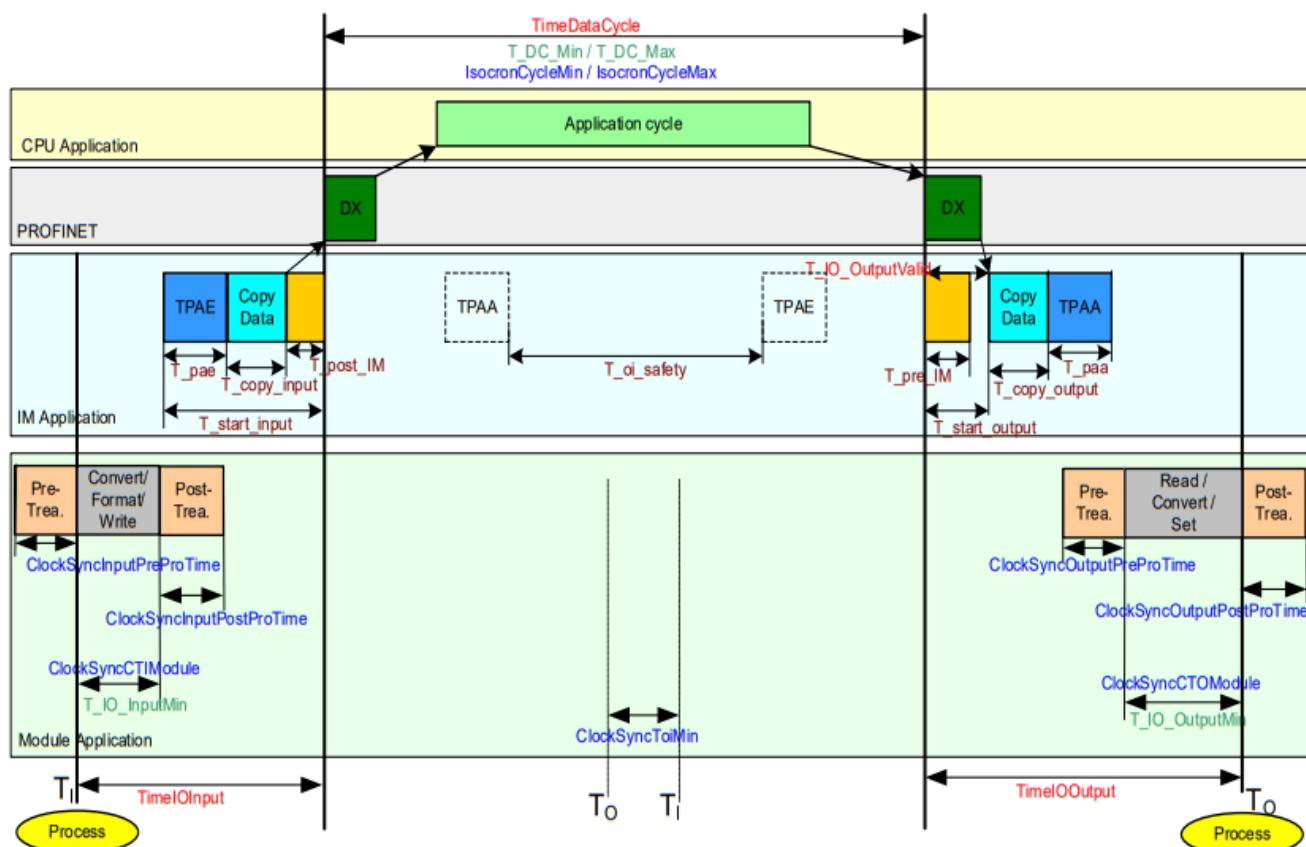
5.4.2.1 Isochronous implementation firmware V1.0

The isochronous mode is a well-defined mechanism. The following features will be supported by the module:

- T_o :
At the timepoint T_o , all output data of the submodule are available to the user logic (CTRL_IF) – the transfer duration from the PBus+ RAM to the logic corresponds to the time ClockSyncCTOModule.
- T_i :
Starting at the timepoint T_i , all input data of the submodule (FB_IF) are transferred to the backplane bus interface, the duration of the transfer between system-logic and PBus+ RAM corresponds to the time ClockSyncCTIModule.

The firmware development must measure the duration of its routines to determine the values for the MDD. MDD values are needed to give TIA-Portal the possibility to calculate the overall isochronous times (T_{dc} , T_i , T_o) for an isochronous rack.

Even in freewheeling mode, the module will run “isochronously” with self-defined values for T_{dc} , T_i , T_o (see table below).



	FASTBig
	Firmware V1.0
TDCMAX	128 (* 31,25µs)
TDCMIN	8 (* 31,25µs)
MDD (V17 HSP and V18+)	
ClocksycnCTIModule	50µs
ClocksycnInputPostProTime	0µs
ClockSyncCTOModule	50µs
ClockSyncOutputPreProTime	50µs(*)
IsocronCycleMin	250µs
IsocronCycleMax	4000µs
Firmware internal	
ClocksycnCTIModule	32µs
ClocksycnInputPostProtime	18µs
ClockSyncCTOModule	30µs
ClockSyncOutputPreProTime	70µs
IsocronCycleMin	250µs
IsocronCycleMax	4000µs
TDC freewheeling	375µs
TI freewheeling	275µs (TDC-100µs)
TO freewheeling	100µs

Table 1: Isochronous timing table V1.0

(*) This value is missing in V18 (value is than 0), therefore it has to be ensured in a V18 project that no TO less than 100us is setup.

Note that the firmware is using $30\mu\text{s}$ for CTI and $30\mu\text{s}$ for CTO, therefore the validation checks will only fail if TI is less than $30\mu\text{s}$ or if TO is bigger than $30\mu\text{s}$.

The following picture is explaining how the module is acting during one TDC cycle:

2 hardware Timers are used to trigger the input and output interrupt routines with high priority: TIMER1 is used for input_isr and TIMER2 for output_isr. This priority in Adonis allows the interruption of any running firmware without latency except fault handlers (HardFault or BusFault). In thread context, the comparator5 (comparator4 for V2.0) set to the time Ti is used to trigger the setting of both TIMER1&2 depending on the remaining time to be counted down (the comparator 5/4 is starting an interrupt routine firing an event to start the high-event-thread).

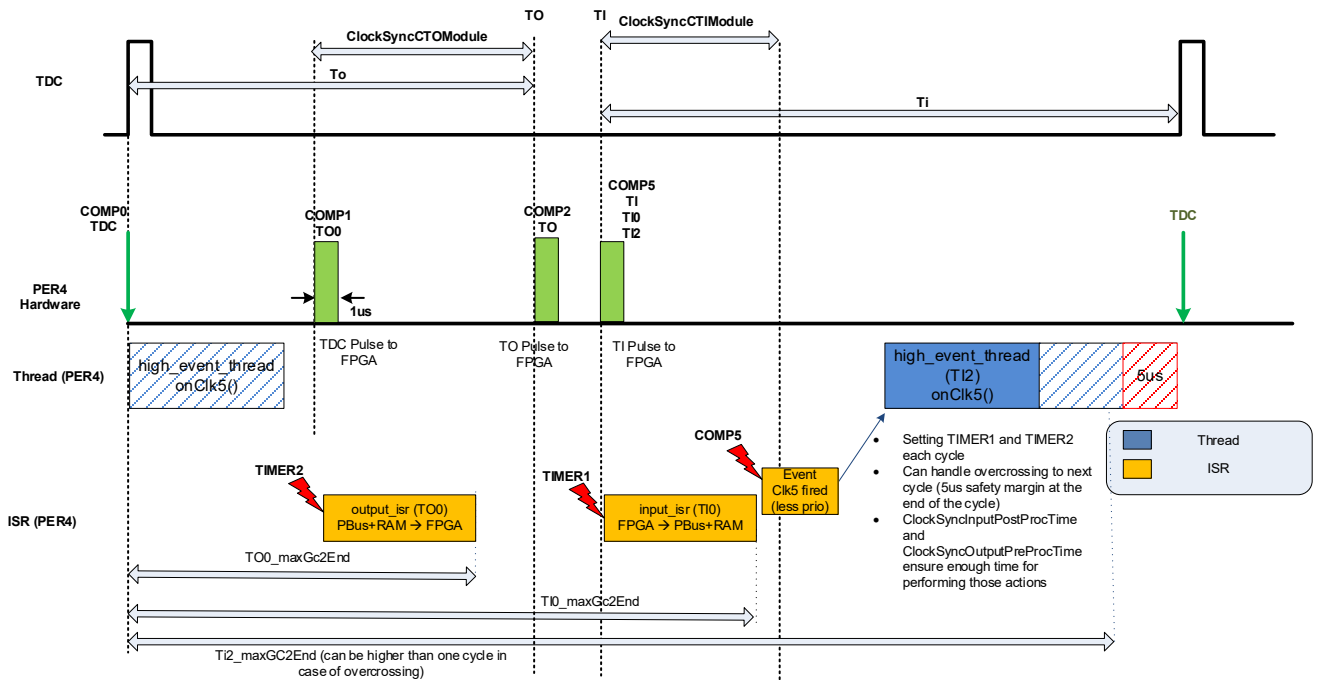


Figure 12: Internal isochronous tasks firmware V1.0

5.4.2.2 Isochronous implementation firmware V2.0

In the firmware V2.0, a parametrizable time is added to give the user application a chance to process the input and output data on time, the new definitions are the following:

- $T_{0\text{-LATCH}}$: At the timepoint To-LATCH, all output data of the submodule are available to the user logic (CTRL_IF), the transfer of the output data from the PBus+ RAM to the system logic will be started upfront. The user logic can start the processing of the output data and must be finished by To.
- T_o :
At the timepoint To, the user application can trigger events at the process IOs
- T_i :
The user application can use this timepoint to start activities on process side (for example triggering the freezing of an encoder value). The transfer of the data from the process side to the feedback interface has to be finished by the time Ti-LATCH.
- $T_{i\text{-LATCH}}$:
At the timepoint Ti-LATCH, all input data of the submodule (FB_IF) are transferred to the backplane bus interface.

The firmware development must measure the duration of its routines to determine the values for the MDD. MDD values are needed to give TIA-Portal the possibility to calculate the overall isochronous times (T_{dc} , T_i , T_o) for an isochronous rack.

Even in freewheeling mode, the module will run “isochronously” with self-defined values for T_{dc} , T_i , T_o (see table below). In freewheeling mode, the parameters UserProcessingTimeInput and UserProcessingTimeOutput will be ignored and internally set to the default value of 100us.

	FASTBig	FASTBigIso
	Firmware V1.0	Firmware V2.0
TDCMAX	128 (* 31,25µs)	128 (* 31,25µs)
TDCMIN	8 (* 31,25µs)	8 (* 31,25µs)
MDD (V17 HSP and V18+)		
ClocksycnCTIModule	50µs	25µs + UserProcessingTimeInput
ClocksycnInputPostProTime	0µs	50µs
ClockSyncCTOModule	50µs	25µs + UserProcessingTimeOutput
ClockSyncOutputPreProTime	50µs(*)	0µs
IsochronCycleMin	250µs	250µs
IsochronCycleMax	4000µs	4000µs
Firmware internal		
ClocksycnCTIModule	32µs	Used: 25µs+UserProcessingTimeInput Measured: 13µs+UserProcessingTimeInput
ClocksycnInputPostProtime	18µs	Measured: 40µs
ClockSyncCTOModule	30µs	Used: 25µs+UserProcessingTimeOutput Measured: 7µs+ UserProcessingTimeInput
ClockSyncOutputPreProTime	70µs	0µs
IsochronCycleMin	250µs	250µs
IsochronCycleMax	4000µs	4000µs
TDC freewheeling	375µs	375µs
TI freewheeling	275µs (TDC-100µs)	275µs (TDC-100µs)
TO freewheeling	100µs	100µs

Table 2: Isochronous timing table V2.0

(*) This value is missing in V18 (value is than 0), therefore it has to be ensured in a V18 project that no TO less than 100us is setup.

The following picture is explaining how the module is acting during one TDC cycle:

2 hardware Timers are used to trigger the input and output interrupt routines with high priority: TIMER1 is used for input_isr and TIMER2 for output_isr. This priority in Adonis allows the interruption of any running firmware without latency except fault handlers (HardFault or BusFault). In thread context, the comparator4 set to the time T_i is used to trigger the setting of both TIMER1&2 depending on the remaining time to be counted down (the comparator4 is starting an interrupt routine firing an event to start the high-event-thread).

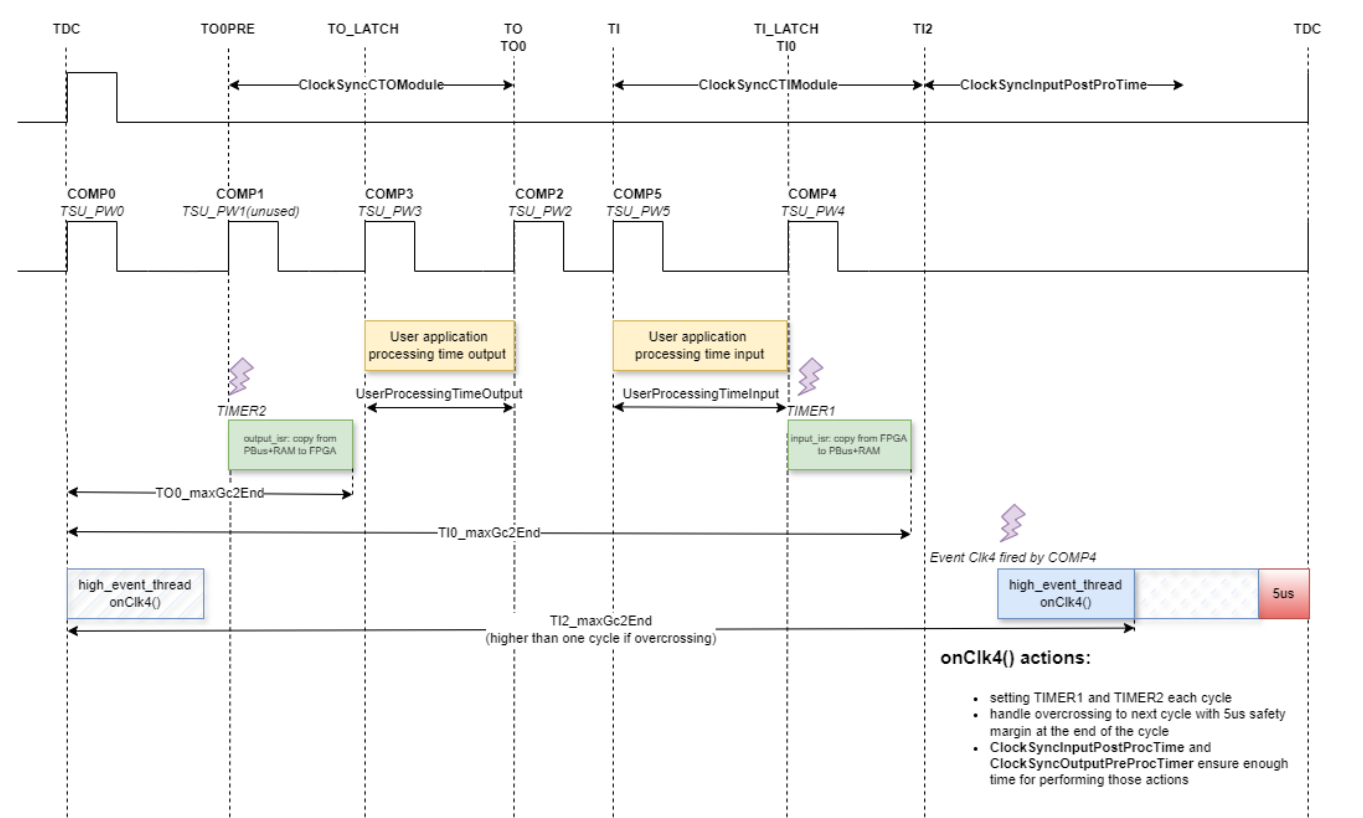


Figure 13: Internal isochronous tasks firmware V2.0

5.4.3 Validation of the isochronous record

The following table is extracted from the SPH “Part 4: Isochronous & High speed IO Modules”. The described checks must be implemented in the backend firmware.

Off set	Name	Type	Values	Meaning	Check in Module
0	BlockType	U32	0x0204	Isochronous Mode Data	=
4	BlockLength	U32	0x001C	Length without Block-Type and BlockLength	=
8	BlockVersionHigh	U8	0x01		=
9	BlockVersionLow	U8	0x00		=
10	Padding	U16	0x0000		No
12	SlotNumber	U16	0x0000 ..0x7FFF		No
14	SubSlotNumber	U16	0x0001 ..0x7FFF		Yes
16	ControllerApplica- tionCycleFactor (CACF)	U16	0x0001 ..0x0400	Reduction in IO-Controller (for checking of sign-of- life)	No
18	TimeDataCycle	U16	0x0001 ..0x0080	Time base 31,25usec Cycle: 31,25usec ..4 ms	$\geq T_DC_Min$ & $\leq T_DC_Max$
20	TimeIOInput	U32	0 ... 4 000 000	Time base is ns	$\geq T_IO_InputMin /$ $ClockSyncCTIModule$ & $\leq 4 000 00$
24	TimeIOOutput	U32	0 ... 4 000 000	Time base is ns	$\geq T_IO_OutputMin /$ $ClockSyncCTOModule$ & $\leq 4 000 00$
28	TimeIOInputValid	U32	1 ... 32 000 000	Time base is ns	No
32	TimeIOOutputValid	U32	1 ... 32 000 000	Time base is ns	No

Table 3: Isochronous record content validation

The validation of the isochronous timing is done at reception of the timing data record (0x8030) and at the end of the parametrization record (128) validation. If the timing cannot be supported, the parametrization record will be rejected with the error "Isochronous Fault" (see parametrization record validation).

➔ At reception of the isochronous record:

- TimeDataCycle is checked against the minimal and maximal cycle time
- TimeIOInput is checked against the CTI time without user application input processing time
- TimeIOOutput is checked against the CTO time without user application output processing time

Hint for the implementation: the checking mechanism is already implemented in the function *writeIsochronousModeData* using the structure *IO_FWK_SM_IDS_ISOCHRONOUS_MODE_MIN_MAX_TABLE*.

➔ At reception of the parametrization record:

- UserProcessingTimeInput must fit in TimeIOInput (see formula in Table 2: Isochronous timing table V2.0
-)
- UserProcessingTimeOutput must fit in TimeIOOutput (see formula in Table 2: Isochronous timing table V2.0
-)

6 Engineering

6.1 Engineering using GSDML

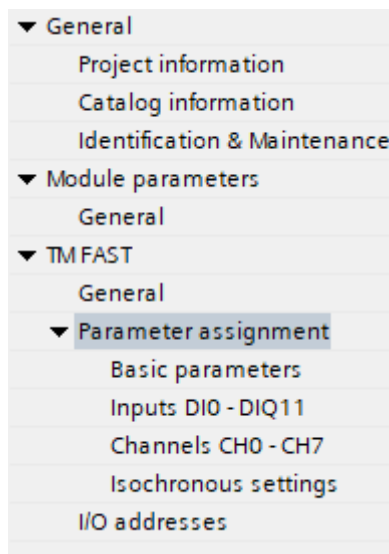
The TM FAST module can be used in a decentral configuration (for example with a non-Siemens CPU) using the GSDML which will be delivered in the GSDML Sharepoint ([Link](#)).
The GSDML will have to support both module IDs. The support of the new DS128 record version is not required by the GSDML as no isochronous mode is possible over GSDML.

6.2 Engineering using TIA HSP/MDD

For TIA Portal V20 TM FAST will be contained in HSP0368 and TM FAST will be fully integrated to TIA V21.
The V2.0 firmware version will be contained in the HSP0368 for V20 and integrated to TIA V21.

6.2.1 Parametrization

Definition for the menu structure in the Properties window. The menu items 'General' and 'Module parameters' are defined identically to any other module. This chapter only describes the UI design for the parametrization. Values and functions of parameters are described in chapter 4.8.1.2



6.2.1.1 TM FAST parameters

6.2.1.1.1 Basic parameters

Basic parameters contain settings for Diagnostic interrupt, Reaction to CPU Stop and TMFASTApplication settings.

> Basic parameters

Diagnostics

☐ Enable diagnostic interrupt on wire break

☐ Digital output error diagnostic interrupt

☐ Load-voltage diagnostic interrupt

TM FAST application

☒ Load TM FAST application at startup

☐ Always activate TM FAST application

Reaction to CPU STOP: Disable all outputs

6.2.1.1.2 Channels DIQ0-11

Input delay settings for DI/Q channels. Menu is identical for every channel.

> Inputs DI0 - DI11

Input delay DI0:	0.1ms
Input delay DI1:	0.001ms
Input delay DI2:	0.003ms
Input delay DI3:	0.05ms
Input delay DI4:	0.1ms
Input delay DI5:	0.4ms
Input delay DI6:	0.8ms
Input delay DI7:	1.6ms
Input delay DI8:	3.2ms
Input delay DI9:	12.8ms
Input delay DI10:	20ms
Input delay DI11:	From TM FAST application

6.2.1.1.3 Channels CH0-7

Rx filter time settings for RS485 channels. Channels 0, 1 and 4 also contain Bus termination checkbox.

> Channels CH0 - CH7

CH0

Input delay: 100µs

☐ Bus termination

CH1

Input delay: 100µs

☐ Bus termination

CH2

Input delay: 100µs

CH3

Input delay: 10µs

CH4

Input delay: 100µs

0.5µs
1µs
2µs
5µs
10µs
20µs
50µs
100µs
200µs
500µs
1000µs
From TM FAST application

6.2.1.1.4 Isochronous settings

Isochronous settings can be applied by setting the user processing time input and the user processing time output. The range lies between 0..4000µs.

> Isochronous settings

User processing time input: 10 µs

User processing time output: 10 µs

i Value range: [0..4000]. **X**

6.2.1.2 Online & Diagnostics

TM FAST diagnostics contain the standard functionality like any other s7-1500 module. Only change is that the category 'Firmware update' shall be renamed to 'Firmware and application update'. This should help with user guidance when updating either component.

The 'Firmware loader' menu is renamed in a similar way to 'Firmware and application loader'.

Firmware and application update

Online data

Article number: 6ES7 554-1AA00-0AB0

Firmware: B 10.0.2

Name: TMFAST_1

Rack: 0

Slot: 2

Firmware and application loader

Firmware file: D:\gitls7p.tmls7p.modules\bin\MP_FAST_1TIAFWUPDATE.575\6ES7 554-1AA00-0AB0 B 10.00.02.UPD

Firmware version: B 10.0.2

Suitable for modules with:

Article number	Firmware version and higher
6ES7 554-1AA00-0AB0	V 1.0.0

Status:

☐ Run firmware after update

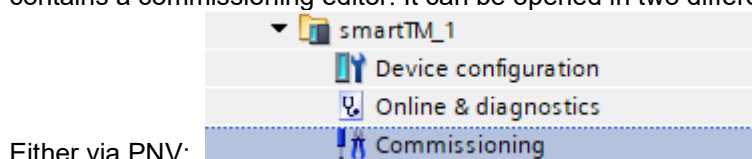
Run update

The option “Run firmware after update” can be checked or not (difference to all other MP modules in TIA). For example, it is possible to only download the firmware or the TMFASTApplication without activating it. Remember that the activation of the new application is depending on the parametrization (see parametrization chapter). Recommendation to the customer added in TIA V21: "Leave 'Run firmware after update' unchecked and refer to the manual for information on how to activate the TMFAST application."

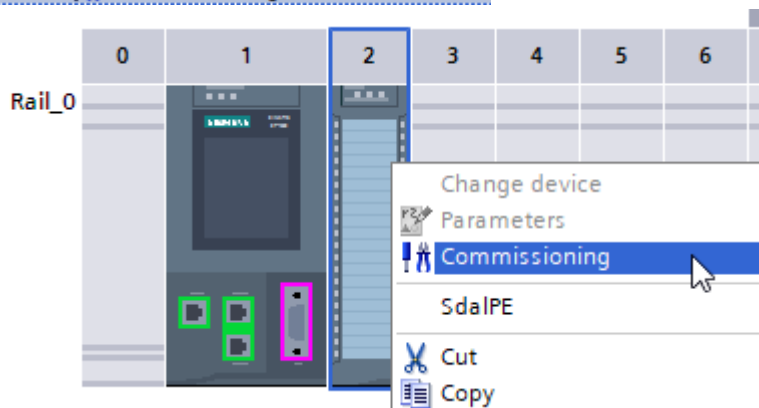
https://jupiter.tfs.siemens.net/tfs/TIA/TIA%20Portal/_workitems/edit/3796176

6.2.1.3 Commissioning

To support customers in the development of TMFASTApplications the TIA Portal integration of the TM FAST contains a commissioning editor. It can be opened in two different ways.



Either via PNV:



Or via context menu:

6.2.1.3.1 Commissioning functions

This chapter describes how TM FAST records can be handled with TIA Portal UI. For a description of the records refer to the chapters 4.7.2 and 4.5.

LTMFAST_ControlREC

This view supports handling of TMFASTControlREC (record 100). Current status is displayed in the upper block. The editor reads the current data once when going online.

There is no cyclic reading of rec 100. If the user performs and action outside this editor that causes changes the refresh must be done manually by clicking 'Read status'. This is only the case when the editor was online during these actions. Download of a different TMFASTApplication would be such a user action.

Under FASTControlREC command the user can select one of the defined commands and send it to the module. Status will be refreshed automatically after the write operation.

TM FAST User Records

To test TMFASTUserREC behavior the possibility to read / write these is available in the commissioning editor. Depending on the length values that are read from the TMFASTControlREC the corresponding number of DWord fields is displayed in the menu TM FAST User Records. Record 102 can be read and record 101 can be written.

There is no cyclic reading or writing of the user records. Any change must be written by clicking the send button. After sending the TMFASTUserWriteREC the TMFASTUserReadREC will be read automatically.

The screenshot displays the LTMFAST ControlREC interface. On the left, a tree view shows the following structure:

- LTMFAST_ControlREC
 - TM FAST User Records (selected)
 - LTMFAST_UserReadREC
 - LTMFAST_UserWriteREC

The main area is divided into two sections:

TM FAST User Records

LTMFAST_UserReadREC

DWord 0:

Last error code:

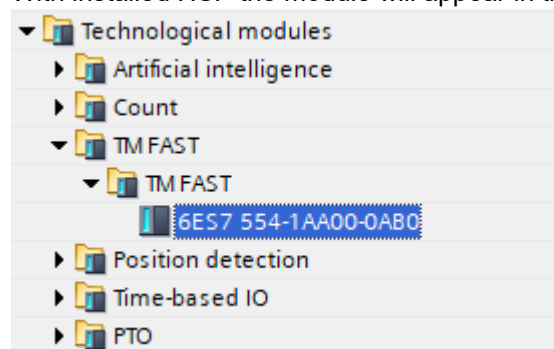
LTMFAST_UserWriteREC

DWord 0:

Last error code:

6.2.2 Hardware catalogue

With installed HSP the module will appear in the hardware catalog of ET200MP and CPU1500 stations like this.



6.2.3 Tooltips

Tooltip IDs are defined in the picture below.

Basic parameters

Diagnostics

Enable diagnostic interrupt on wire break

Digital output error diagnostic interrupt

Load-voltage diagnostic interrupt

smartTM application

Load smartTM application at startup

Always activate smartTM application

Reaction to CPU STOP:

Disable all outputs

Channels CH0 - CH3

CH0

Input delay: 100µs

CH1

Input delay: 100µs

Bus termination

CH2

Input delay: 100µs

0.5µs

1µs

2µs

5µs

CH3

Input delay: 100µs

20µs

50µs

100µs

200µs

500µs

1000µs

CH4

Input delay: From smartTM application

Inputs DI0 - DIQ11

Input delay DI0: 0.1ms

Input delay DI1: 0.001ms

Input delay DI2: 0.003ms

Input delay DI3: 0.05ms

Input delay DI4: 0.1ms

Input delay DI5: 0.4ms

Input delay DI6: 0.8ms

Input delay DI7: 1.6ms

Input delay DI8: 3.2ms

Input delay DI9: 12.8ms

Input delay DI10: 20ms

Input delay DI11: From smartTM application

15386

15387

15388

15363

15364

15389

15366

15390

15385

Template A5E00785367A (14.05.24)
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Feature Specification

LTMFAST_ControlREC TM FAST User Records LTMFAST_UserReadREC LTMFAST_UserWriteREC	LTMFAST_ControlREC TM FAST module state FPGA state: TmFAST application is loaded and active 15374 Last error code: No error 15375 Debug interface state: Debug interface is deactivated 15394 Internal temperature: 32.5 °C 15380 Module is ready for a new command 15395 Read status
	LTMFAST_ControlREC command Command: Load TmFAST application from flash memory to FPGA 15383 Send command
	Currently loaded TM FAST Application Application ID: TEST 4 15376 User logic version: B 01.02.43 15378 LTMFAST_UserWriteREC length: 4 Byte 15381 LTMFAST_UserReadREC length: 4 Byte 15382
	Flash logic status Application ID: TEST4 15391 User logic version: B 01.07.20 15392 Time stamp: Tuesday, July 19, 2022 09:20 15393
LTMFAST_ControlREC TM FAST User Records LTMFAST_UserReadREC LTMFAST_UserWriteREC	TM FAST User Records LTMFAST_UserReadREC DWord 0: 0 Hex 15384 Read record Last error code: No error 15396 LTMFAST_UserWriteREC DWord 0: 0 Hex 15385 Write record Last error code: No error 15397

Neue Tooltips für die V2.0:

User processing time input:	10	µs	15367
User processing time output:	10	µs	15368
Value range: [0..4000].			X

6.3 Engineering for Profibus

For using the module with Profibus, the following configurations are theoretically possible, the decision which configurations will be supported is to be found at end of this chapter:

- DPV0

By using DPV0 you don't have any possibility to send a data set. The only way to distribute the parameter data is to

use the PRM-telegram. With this telegram there are only 220 Octets usable for the whole rack. At this reason the data set 245 (this data set will be part of the PRM-telegram) has to be as small as possible.

- GSD with non-Siemens CPU:
DPV0 is only possible with a lot of limitation. Do we really need DPV0
- GSD with Siemens CPU:
Makes no sense cause Siemens CPUs support the startup data set mechanism (this CPUs will send data set 128 in DPV1).
- MDD with Siemens CPU:
ditto.
- DPV1
 - GSD with non-Siemens CPU:
Data set 245 is needed for the PRM-telegram. But the customer can send the data set 128 after startup to the module and reach the whole functionality of the module.
 - GSD with Siemens CPU:
Data set 245 is needed for the PRM-telegram. But the customer can send the data set 128 after startup to the module and reach the whole functionality of the module.
 - MDD with Siemens CPU:
This is the best way. By using a SIMATIC CPU you will get the automatic startup mechanism → the CPU will send the data set 128 during startup to the module.

According to [SPH \(siemens.com\)](https://www.siemens.com), the parameter record length for Profibus shall not be longer than 16 bytes, otherwise a limitation of the pluggable modules is necessary, therefore a short parametrization record (index 245) has to be implemented.

Add Profibus config telegram

Decision for the TM FAST module supported Profibus configuration:

- DPV0 will not be supported
- DPV1 will be supported but only with MDD – therefore no support of the data record 245 will be implemented in the firmware

6.4 TM FAST TIA Library (LTMFAST)

6.4.1 Introduction

In TIA Portal, the library LTMFAST can be used to facilitate the communication between the S7 user-program and the S7-1500 TM FAST module by providing FBs (Function Blocks) and data types.

LTMFAST library is using specific instructions like the data type ARRAY[*] and the block FileReadC to access the memory card. Therefore, the LTMFAST library can only be used on the following CPUs:

- ➔ S7-1200 series as of firmware version ≥ 4.2 (**only without LTMFAST_AppDownload block**)
- ➔ S7-1500 series as of firmware version ≥ 2.5
- ➔ S7-1500 series as of firmware version ≥ 2.0 (**only without LTMFAST_AppDownload block**)

6.4.2 LTMFAST library content

6.4.2.1 LTMFASTAppDownload

This block can download the userlogic (TMFAST application) from SMC or from DB.

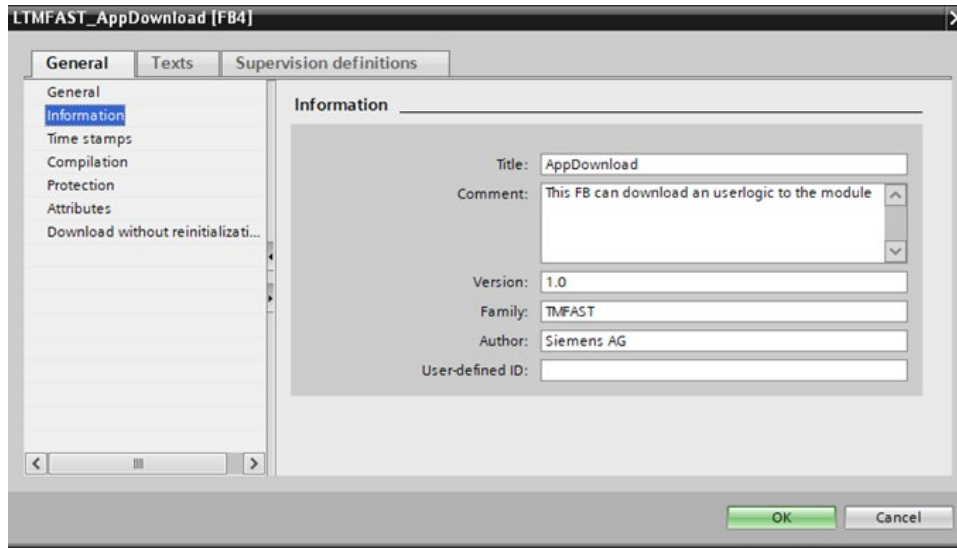
6.4.2.1.1 FB Information

This block can be used with the S71500 and OpenController.

This block can only be used with the S71500 with firmware version ≥ 2.5 .

This block is protected.

The attributes IEC check and optimized block access are active.

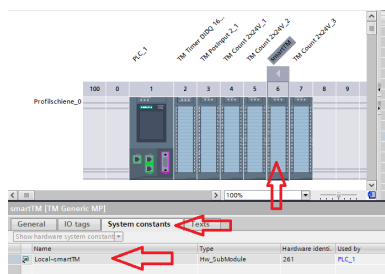


6.4.2.1.2 Block interface

```
"LTMFAST_AppDownload_DB"(req          := false,
                           HW_ID       := 0,
                           filenameSMC := '',
                           userlogicDB := 0,
                           done        => _bool_out_,
                           busy        => _bool_out_,
                           error       => _bool_out_,
                           status      => _word_out_);
```

6.4.2.1.3 FB parameters

- req: Edge triggered boolean input to start the block
- HW_ID: HW_ID is the hardware identifier of the TMFAST Module.



- filenameSMC: the filename of the userlogic you want to load from SMC
example of the path: 'UserFiles/Userlogic.upd'
UPD file with the userlogic can be created with MFCT
- userlogicDB: the number of the DB which contains the userlogic. Set this value to 0 if you want to load the userlogic from a file on the SMC instead of a DB
SCL source for the userlogicDB can be created with MFCT

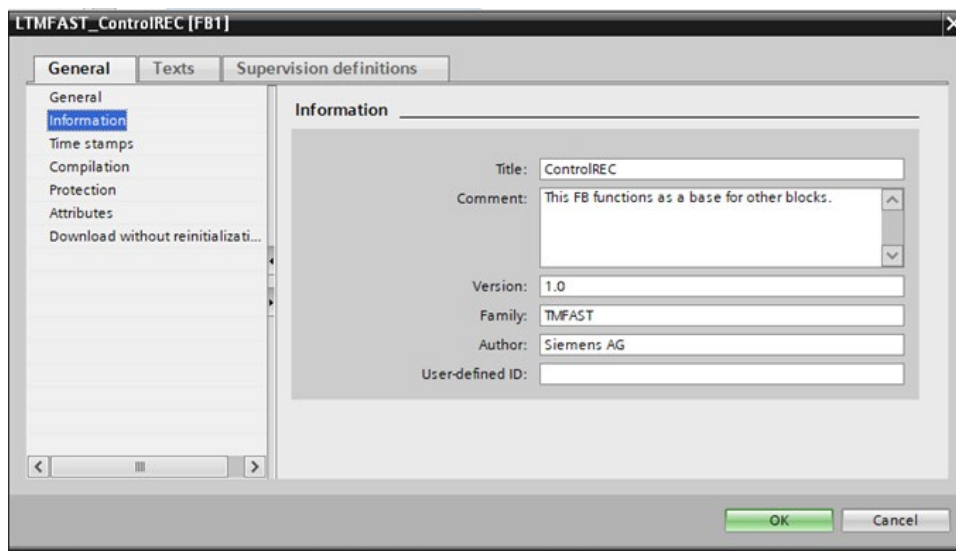
- done: Returns true for one program cycle after the block is done
- busy: Returns true while the block is active
- error: Returns true for one program cycle in case of error
- status: In case of an error this gives an error code for one program cycle
 - W#16#7000: No job running
 - W#16#7001: Initialization running
 - W#16#7002: Download user data
 - W#16#8001: Error WRREC RDREC
 - W#16#8002: Error during reading user data from DB
 - W#16#8003: Error during reading user data from SMC
 - W#16#8004: Error at handling of user data
 - W#16#8005: Error Userlogic image defect
 - W#16#8006: Error wrong State

6.4.2.2 LTMFASTControlREC

This block can control the states of TMFastApplication.

6.4.2.2.1 FB information

This block can be used with the S71500, OpenController and S71200.
This block can only be used with the S71500 with firmware version ≥ 2.0
This block can only be used with the S71200 with firmware version ≥ 4.2
This block is not protected.
The attributes IEC check and optimized block access are active.



6.4.2.2.2 Block interface

```
"LTMFAST_ControlREC_DB"( req           := false,
                          HW_ID         := 0,
                          command       := 16#0,
                          FPGAState    => _byte_out_,
                          applicationID => _array_of_char_out_,
                          userLogicVersion => _ltmfast_typedlogicversion_out_,
                          debugInterfaceState => _bool_out_,
                          done          => _bool_out_,
                          busy          => _bool_out_,
                          status        => _dword_out_,
```

error ==> _bool_out_);

6.4.2.2.3 FB parameters

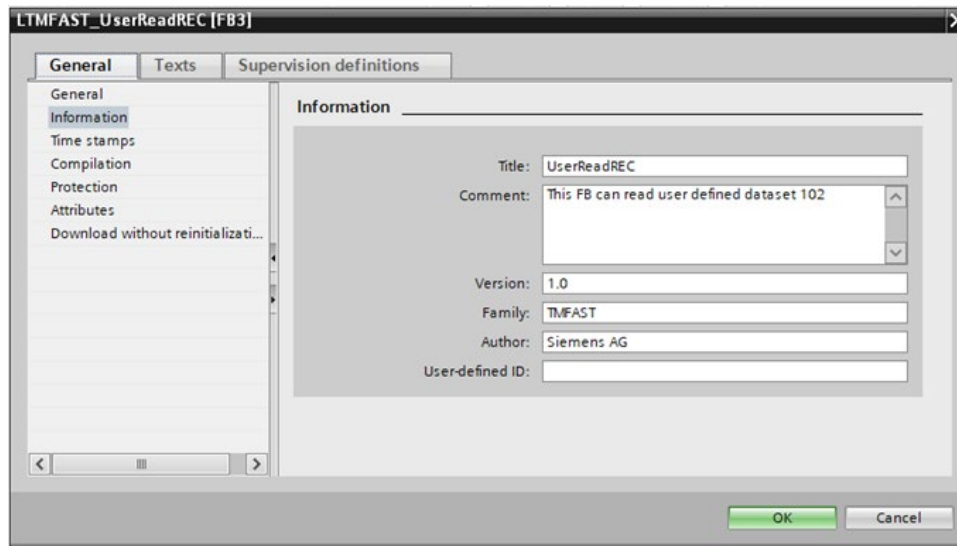
- req: Edge triggered boolean input to start the block
- HW_ID: HW_ID is the hardware identifier of the TMFAST Module.
- Command:
 - o 0: Only read module state
 - o 1: Load TMFASTApplication from flash
 - o 2: Activate a loaded TMFASTApplication
 - o 3: Clear the TMFASTApplication from FPGA
 - o 4: Delete the TMFASTApplication from flash
 - o 5: Activate debug interface
 - o 6: Deactivate debug interface
- FPGAState:
 - o 0: TMFASTApplication is cleared
 - o 1: TMFASTApplication is loaded but not active
 - o 2: TMFASTApplication is loaded and active
- applicationID:
 - o 8 characters with the ID of the loaded TMFASTApplication
- userLogicVersion:
 - o Version of the loaded TMFASTApplication, example "R01.02.03"
- debugInterfaceState:
 - o 0: FPGA Debug interface is deactivated
 - o 1: FPGA Debug interface is activated
- done: Returns true for one program cycle after the block is done
- busy: Returns true while the block is active
- error: Returns true for one program cycle in case of error
- status: In case of an error this gives an error code for one program cycle
 - o W#16#7001 WRREC in progress
 - o W#16#7002 RDREC in progress
 - o W#16#8001 Input addresses wrong length
 - o W#16#8002 Output addresses wrong length
 - o W#16#8003 Error in RDREC
 - o W#16#8004 Error in WRREC
 - o W#16#8005 Error in RD_ADDR

6.4.2.3 LTMFASTUserReadREC

This block can read user defined dataset 102.

6.4.2.3.1 FB information

This block can be used with the S71500, OpenController and S71200.
This block is not protected.
The attributes IEC check and optimized block access are active.



6.4.2.3.2 Block interface

```
"LTMFAST_UserReadREC_DB"(req      := false,
                           len      => _uint_out_,
                           busy     => _bool_out_,
                           done     => _bool_out_,
                           error    => _bool_out_,
                           status   => _word_out_,
                           userDataContent := _array_of_dword_inout_,
                           TMFASTControlREC := _param_fb_inout_);
```

6.4.2.3.3 FB parameters

- req: Edge triggered boolean input to start the block.
- len: Returns the length of the read user data set in bytes.
- busy: Returns true while the block is active.
- done: Returns true for one program cycle after the block is done.
- error: Returns true for one program cycle in case of error.
- status: In case of an error this gives an error code for one program cycle.
 - o W#16#7001: Initialization running
 - o W#16#7002: Reading user data
 - o W#16#8001: Error during initialization
 - o W#16#8002: Error during reading user data
- userDataContent: Returns an array of DWords containing the read user defined dataset.
- TMFASTControlREC: has to be assigned to the instance of LTMFAST ControlREC that is connected to the used TMFAST module.

6.4.2.4 LTMFASTUserWriteREC

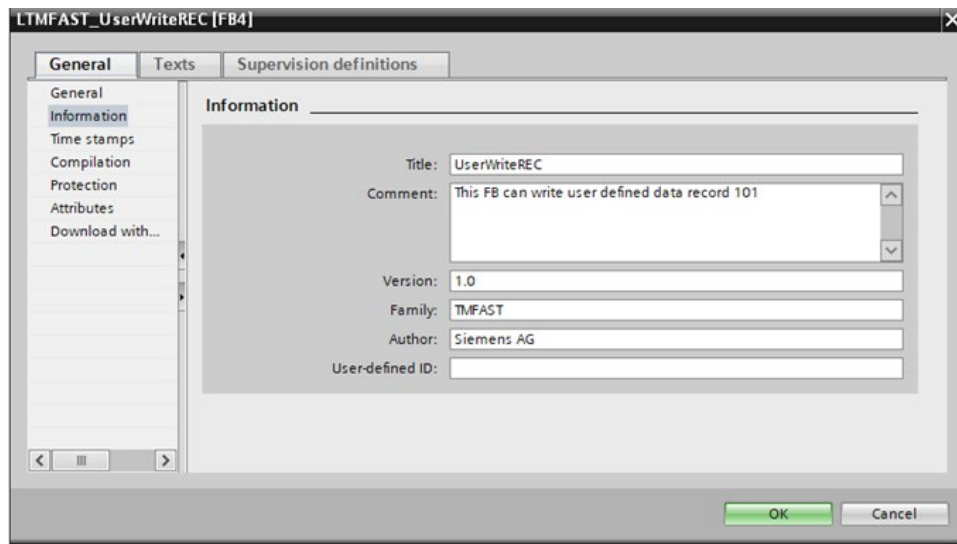
This block can read user defined dataset 101.

6.4.2.4.1 FB information

This block can be used with the S71500, OpenController and S71200.

This block is not protected.

The attributes IEC check and optimized block access are active.



6.4.2.4.2 Block interface

```
"LTMFAST_UserWriteREC_DB"(req           := false,
                                activateRead := false,
                                busy          => _bool_out_,
                                done          => _bool_out_,
                                error         => _bool_out_,
                                status        => _word_out_,
                                userDataContent := _array_of_dword_inout_,
                                TMFASTControlREC := _param_fb_inout_);
```

6.4.2.4.3 FB parameters

- req: Edge triggered boolean input to start the block.
- activeRead:
 - False: Block writes the data record 101 with data that is parameterized at UserDataContent.
 - True: Block reads data record 101 from the module and writes its content into UserDataContent.
- busy: Returns true while the block is active.
- done: Returns true for one program cycle after the block is done.
- error: Returns true for one program cycle in case of error.
- status: In case of an error this gives an error code for one program cycle.
 - W#16#7001: Initialization running
 - W#16#7002: Reading user data
 - W#16#7003: Writing user data
 - W#16#8001: Error during initialization
 - W#16#8002: Error during reading user data
 - W#16#8003: Error during writing user data
- userDataContent: Returns an array of DWords containing the read user defined dataset.
- TMFASTControlREC: got to be assign the instance of LTMFAST ControlREC that is connected to the used TMFAST modul.

6.4.2.5 User data types (UDTs)

The library also contains 4 UDTs:

- LTMFAST_typeAppDownloadData
- LTMFAST_typeControlREC_RD
- LTMFAST_typeControlREC_WR

- LTMFAST_typeLogicVersion

7 Compatibility with previous versions

This chapter is focused on the compatibility between firmware versions (V1.0 and V2.0) and system logic versions (V1.0 and V2.0)

7.1 Interfaces

The interface between Firmware and system logic is realized over hardware signals (isochronous Tics signals: TSU_PWx) and registers accessed by the firmware over the CtC interface.
In comparison to the version V1.0, the system logic V2.0 is supporting 2 additional Tic signals (TSU_PW3 for T_O_LATCH and TSU_PW4 for T_I_LATCH) and one new register bit in the register TFL_FAST_SYS.MODE called ADVANCED_ISOCHRONOUS. Only if this bit is set by the firmware (by default it is not), the system logic will use the new signals T_O_LATCH and T_I_LATCH.

7.2 Functionality

The following picture is summarizing the different configurations, more details are in the text below.

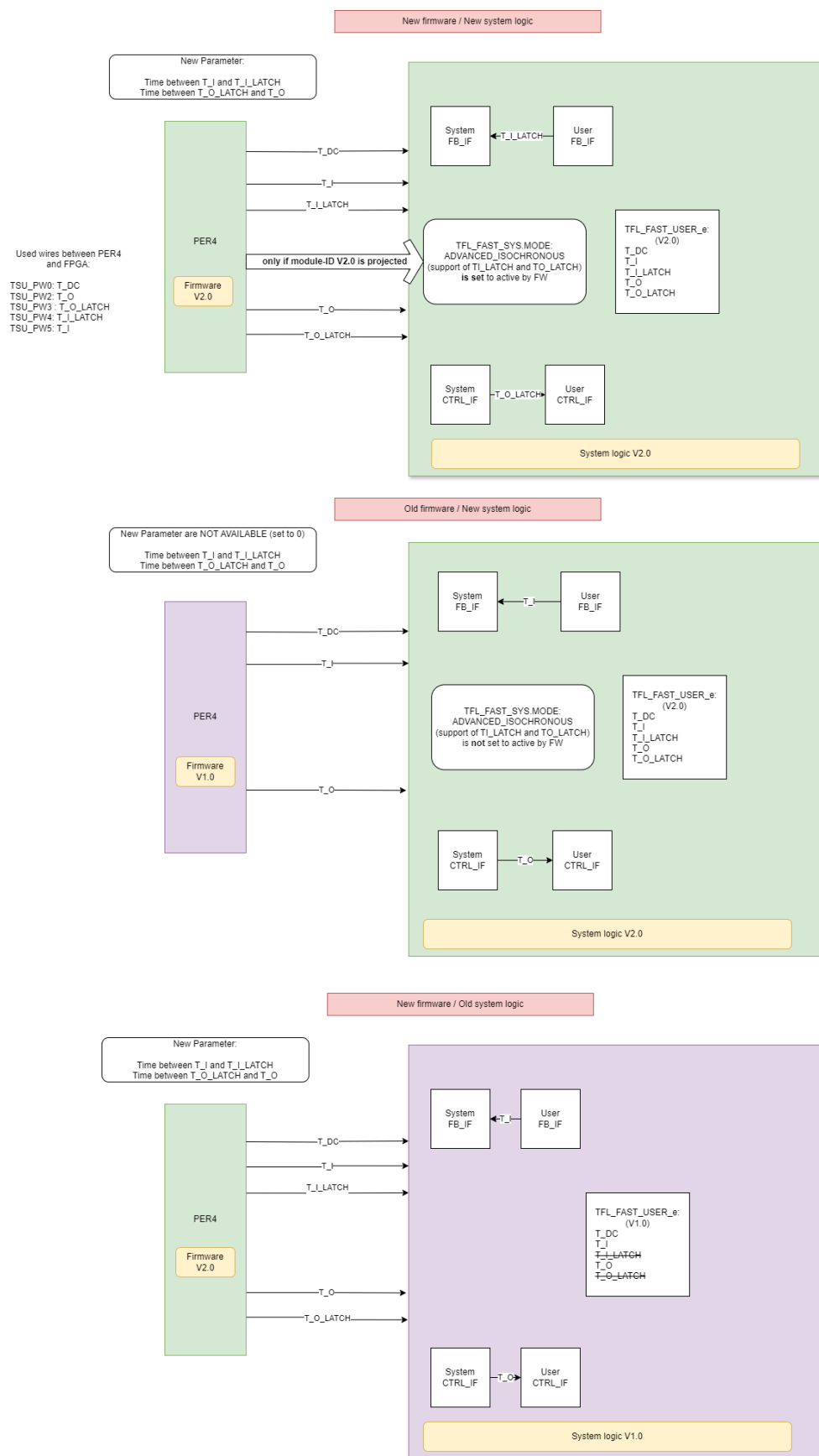


Figure 14: Compatibility cases firmware and system logic

7.2.1.1 *Old firmware V1.0 / new system logic V2.0*

In this configuration the new system logic will work as the old logic and use T_I to latch the feedback interface and T_O to latch the control interface. We will **strongly recommend** for any user downloading the new system logic to update the firmware but will not force him to do so by not rejecting the application.

The firmware V1.0 will deliver the pulses over the following wires to the FPGA:

TSU_PW0: T_DC

TSU_PW2: T_O

TSU_PW5: T_I

7.2.1.2 *New firmware V2.0 / new system logic V2.0*

This recommended configuration supports the extended isochronous mode, the user can set the position of the latch signals for latching the feedback interface and the control interface.

The firmware will deliver the different time pulses to the FPGA using the following wires:

TSU_PW0: T_DC

TSU_PW2: T_O

TSU_PW3: T_O_LATCH

TSU_PW4 : T_I_LATCH

TSU_PW5: T_I

If the old module-ID (V1.0) is projected, the system logic will use T_I and T_O to latch the IO-data. If the new module-ID (V2.0) is projected, the system logic will use the T_I_LATCH to latch the feedback interface of the user logic into the system logic and T_O_LATCH to latch the control interface of the system logic into the user logic.

7.2.1.3 *New firmware V2.0 / old system logic V1.0*

In this configuration, the system logic will ignore T_O_LATCH and T_I_LATCH and continue to work as in the past. There is currently no plan to inform the customer that the parametrization of T_O_LATCH and T_I_LATCH has no effect on the user application. The Tooltips about the new isochronous parameters will have to report that the usage of the system logic V2.0 is a prerequisite to make use of the parameters.

8 Appendix

8.1 Terms/Abbreviations

Term / Abbreviation	Explanation
FPGA	Field Programmable Gate Array

8.2 Referenced Documents

Reference	Document with title, definite identifier and version
Requirements	Requirement Specification Document: Link MindMap with list of detailed list of requirements: Link
BackplaneBoardSchematic	Schematic of the backplane PCB: Link
Cyclone10LPHandbook	Cyclone 10 LP Core Fabric and General Purpose I/Os Handbook: Link
I2C_and_TMP100_sensor_wiki	I2C and TMP 100 sensor description: Link
SPH Security	SPH Security S7p_mod14